

INSTITUTO SUPERIOR TÉCNICO

MASTERS IN MECHANICAL ENGINEERING

OPTIMAL CONTROL LABORATORY PROJECT - PHASE II

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Modeling and Simulation of a Quadcopter

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Introduction

1 Kalman-Bucy Observer

1.1 Sources of Measurement Noise

The measurement noise arises from the physical limitations of the sensors used to obtain each output, whose variances are summarised in Table 1.

Output	Symbol	Variance
Inertial Position X	r_1 (m)	7.3221×10^{-1}
Inertial Position Y	r_2 (m)	4.2821×10^{-1}
Inertial Position Z	r_3 (m)	1.8010×10^{-3}
Roll	θ_1 (rad)	3.4817×10^{-4}
Pitch	θ_2 (rad)	2.6060×10^{-4}
Yaw	θ_3 (rad)	3.8313×10^{-2}
Body Angular Velocity X	ω_1 (rad/s)	1.8271×10^{-6}
Body Angular Velocity Y	ω_2 (rad/s)	4.1449×10^{-6}
Body Angular Velocity Z	ω_3 (rad/s)	7.0775×10^{-6}
Body Linear Velocity X	v_1 (m/s)	2.2190×10^{-3}
Body Linear Velocity Y	v_2 (m/s)	2.2190×10^{-3}
Body Linear Velocity Z	v_3 (m/s)	1.8271×10^{-4}

Table 1: Measurement noise variances σ_M^2 for each output.

The main sources of measurement noise are:

- **Inertial positions** r_1, r_2, r_3 — measured by the external localisation system, subject to limited resolution and signal interference. The much larger variances in r_1 and r_2 relative to r_3 reflect lower horizontal localisation accuracy.
- **Orientation angles** $\theta_1, \theta_2, \theta_3$ **and angular velocities** $\omega_1, \omega_2, \omega_3$ — measured by the onboard inertial sensors, affected by propeller vibrations, sensor bias, and thermal drift. Yaw θ_3 has a higher variance than roll and pitch due to drift accumulation.
- **Linear velocities** v_1, v_2, v_3 — obtained by differentiating position measurements, which amplifies the localisation noise.
- **All channels** — affected by electrical noise from the onboard electronics and by aerodynamic disturbances from the propellers. Additionally, errors in the orientation estimate $\mathbf{R}_n$ propagate into position through $\dot{\mathbf{r}}_n = \mathbf{R}_n \mathbf{v}_n$, coupling orientation and position noise.

1.2 Sources of Process Noise

Process noise models deviations from the true motion model, arising from any phenomenon not captured by the quadcopter's equations of motion. The process noise variances σ_P^2 assumed for each state are summarised in Table 2. The main sources are the following.

State	Symbol	Variance
Inertial Position X	r_1 (m)	9×10^{-1}
Inertial Position Y	r_2 (m)	9×10^{-1}
Inertial Position Z	r_3 (m)	1×10^{-1}
Roll	θ_1 (rad)	3×10^{-1}
Pitch	θ_2 (rad)	2×10^{-1}
Yaw	θ_3 (rad)	1×10^{-1}
Body Angular Velocity X	ω_1 (rad/s)	1×10^{-3}
Body Angular Velocity Y	ω_2 (rad/s)	1×10^{-3}
Body Angular Velocity Z	ω_3 (rad/s)	1×10^{-3}
Body Linear Velocity X	v_1 (m/s)	2×10^{-1}
Body Linear Velocity Y	v_2 (m/s)	2×10^{-1}
Body Linear Velocity Z	v_3 (m/s)	1×10^{-2}

Table 2: Process noise variances σ_P^2 for each state.

- **Unmodelled aerodynamics** — the model assumes propeller thrust is strictly proportional to Ω_i^2 , neglecting blade flapping, air drag, and rotor-to-rotor airflow interactions. These effects cause the real dynamics to deviate from the model predictions.
- **Motor dynamics** — the document assumes motor dynamics are significantly faster than the rigid body dynamics, modelling them as a static gain $\Omega_i^2 = K_{u \rightarrow \Omega^2} u_i$. Any deviation from this assumption, including battery voltage variations affecting motor response, introduces process noise.
- **Rigid body assumption** — the drone is modelled as a perfectly rigid body. In reality, frame vibrations and propeller blade deformations cause small deviations from the predicted motion.
- **Linearization error** — the observer is built on the linearized model, so the mismatch between the linearized and true nonlinear dynamics acts as an additional source of process noise.

1.3 Process and Measurement Noise Covariance Matrices

In the Kalman-Bucy filter, the matrices Q_0 and R_0 represent:

- **Process noise covariance matrix** Q_0 — models the uncertainty in the system dynamics, accounting for unmodelled phenomena that cause the true state to deviate from the model prediction. Its diagonal entries correspond to the process noise variances σ_P^2 of each state.
- **Measurement noise covariance matrix** R_0 — models the uncertainty in the sensor measurements, reflecting the noise corrupting the observed outputs. Its diagonal entries correspond to the measurement noise variances σ_M^2 of each output.

Both matrices are symmetric and positive semi-definite by definition, and are assumed diagonal since the noise sources are considered independent across channels.

Regarding their dimensions, Q_0 is $n \times n$ where n is the number of states, and R_0 is $p \times p$ where p is the number of measured outputs. Since all 12 states are directly measured, both matrices are 12×12 for the full system. When considering each decoupled subsystem individually, the dimensions reduce to:

- **Vertical and Yaw subsystems** — Q_0 and R_0 are 2×2
- **Lateral and Longitudinal subsystems** — Q_0 and R_0 are 4×4

1.4 Optimal Choice for Q_0 and R_0

The optimal choice for Q_0 and R_0 is to set their diagonal entries equal to the true noise variances provided in Tables 2 and 1, respectively:

$$Q_0 = \text{diag}(\sigma_{P,r_1}^2, \sigma_{P,r_2}^2, \sigma_{P,r_3}^2, \sigma_{P,\theta_1}^2, \sigma_{P,\theta_2}^2, \sigma_{P,\theta_3}^2, \sigma_{P,\omega_1}^2, \sigma_{P,\omega_2}^2, \sigma_{P,\omega_3}^2, \sigma_{P,v_1}^2, \sigma_{P,v_2}^2, \sigma_{P,v_3}^2) \quad (1)$$

$$R_0 = \text{diag}(\sigma_{M,r_1}^2, \sigma_{M,r_2}^2, \sigma_{M,r_3}^2, \sigma_{M,\theta_1}^2, \sigma_{M,\theta_2}^2, \sigma_{M,\theta_3}^2, \sigma_{M,\omega_1}^2, \sigma_{M,\omega_2}^2, \sigma_{M,\omega_3}^2, \sigma_{M,v_1}^2, \sigma_{M,v_2}^2, \sigma_{M,v_3}^2) \quad (2)$$

This is the optimal choice because the Kalman-Bucy filter minimises the state estimation error variance only when Q_0 and R_0 exactly reflect the true statistics of the process and measurement noise, respectively. Any deviation from

the true variances results in a suboptimal estimator, as the filter will incorrectly balance its trust between the model prediction and the sensor measurements.

1.5 Observer Validation on the Linearized Model

The Kalman-Bucy observer was validated on the linearized model with the pole-placement regulator from Phase I. The state vector of the full system is defined as:

$$x = \begin{bmatrix} r_3 & \dot{r}_3 & r_2 & \theta_1 & \dot{r}_2 & \dot{\theta}_1 & r_1 & \theta_2 & \dot{r}_1 & \dot{\theta}_2 \end{bmatrix}^\top \quad (3)$$

The system and observer were initialized with deliberately different initial conditions, as shown in Equations 4 and 5, to assess the observer's ability to converge to the true system states from an incorrect initial estimate.

$$x_0^{\text{system}} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & & & & & & & & \end{bmatrix}^\top \quad (4)$$

$$x_0^{\text{observer}} = \begin{bmatrix} 3 & 0 & 3 & 0 & 0 & 0 & 3 & 0 & 0 & 0 \\ 0.5 & 0 & & & & & & & & \end{bmatrix}^\top \quad (5)$$

The reference values used are consistent with those from Phase I:

$$r_1^{\text{ref}} = 2 \text{ m}, \quad r_2^{\text{ref}} = 1 \text{ m}, \quad r_3^{\text{ref}} = 4 \text{ m}, \quad \theta_3^{\text{ref}} = 0.5 \text{ rad} \quad (6)$$

Figure 1 shows the evolution of the key outputs — the positions r_1 , r_2 , r_3 and the yaw angle θ_3 — comparing the noisy measurements y with the Kalman-Bucy estimated states $\hat{x}$.

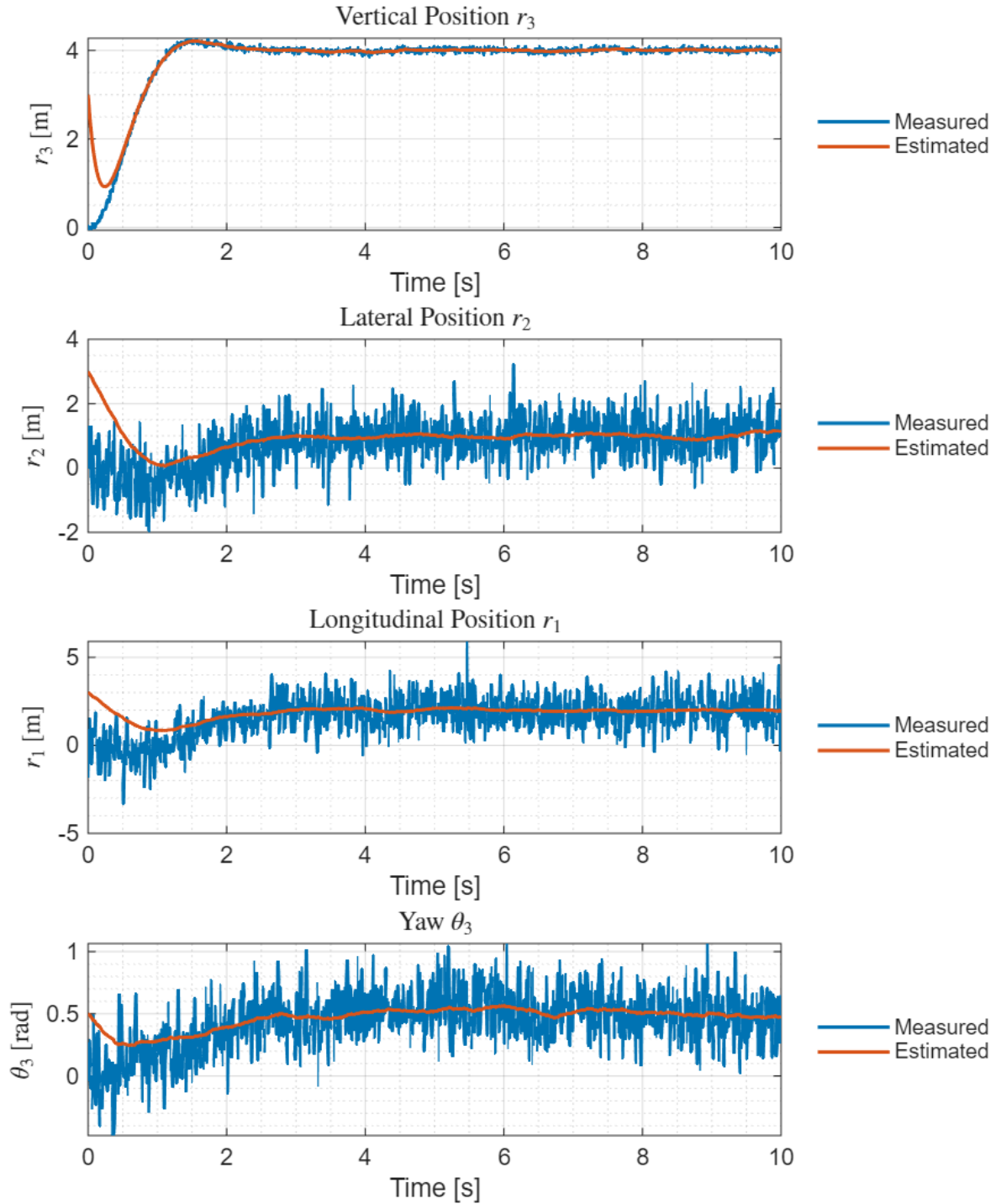
Q5 | Key Outputs: r_1, r_2, r_3, θ_3


Figure 1: Q5 — Kalman-Bucy observer validation on the linearized model. Noisy measurement (blue) vs. estimated state (orange) for the key outputs r_3 , r_2 , r_1 , and θ_3 .

The estimated states converge to the true system states in all four channels. The vertical position r_3 exhibits the fastest and cleanest convergence, consistent with its significantly lower measurement noise variance ($\sigma_{M,r_3}^2 = 1.80 \times 10^{-3} \text{ m}^2$) compared to the horizontal positions. The lateral position r_2 , longitudinal position r_1 , and yaw angle θ_3 show larger measurement noise in the raw sensor signal, reflected in the wider spread of the blue measured signal. In all cases, the

Kalman-Bucy filter successfully smooths the noise, producing a clean estimate that tracks the true trajectory. This confirms that the observer eigenvalues, placed in the left half-plane through the optimal Kalman gain L , guarantee stable convergence of the estimation error to zero.

Despite the system and observer starting from different initial conditions, convergence speed varies across channels, with r_3 correcting almost instantaneously and r_1 taking slightly longer due to its higher measurement noise variance. This reflects the direct influence of R_0 on the Kalman gain magnitude and therefore on the convergence rate of each channel.

It is also worth noting that the smoothness of the estimated signal comes at the cost of some tracking speed — channels with higher measurement noise variance, such as r_1 and r_2 , show a slightly lagged estimate during fast transients. This is an inherent trade-off in the Kalman filter between noise rejection and tracking bandwidth, governed by the relative magnitudes of Q_0 and R_0 .

1.6 Sensitivity Analysis of Q_0

The process noise covariance matrix Q_0 represents the confidence level in the accuracy of the motion model. Five scaling factors were tested — $Q_0 \times 0.01$, $Q_0 \times 0.1$, optimal Q_0 , $Q_0 \times 10$, and $Q_0 \times 100$ — while keeping R_0 fixed at its optimal value. The results are shown in Figures 2–5.

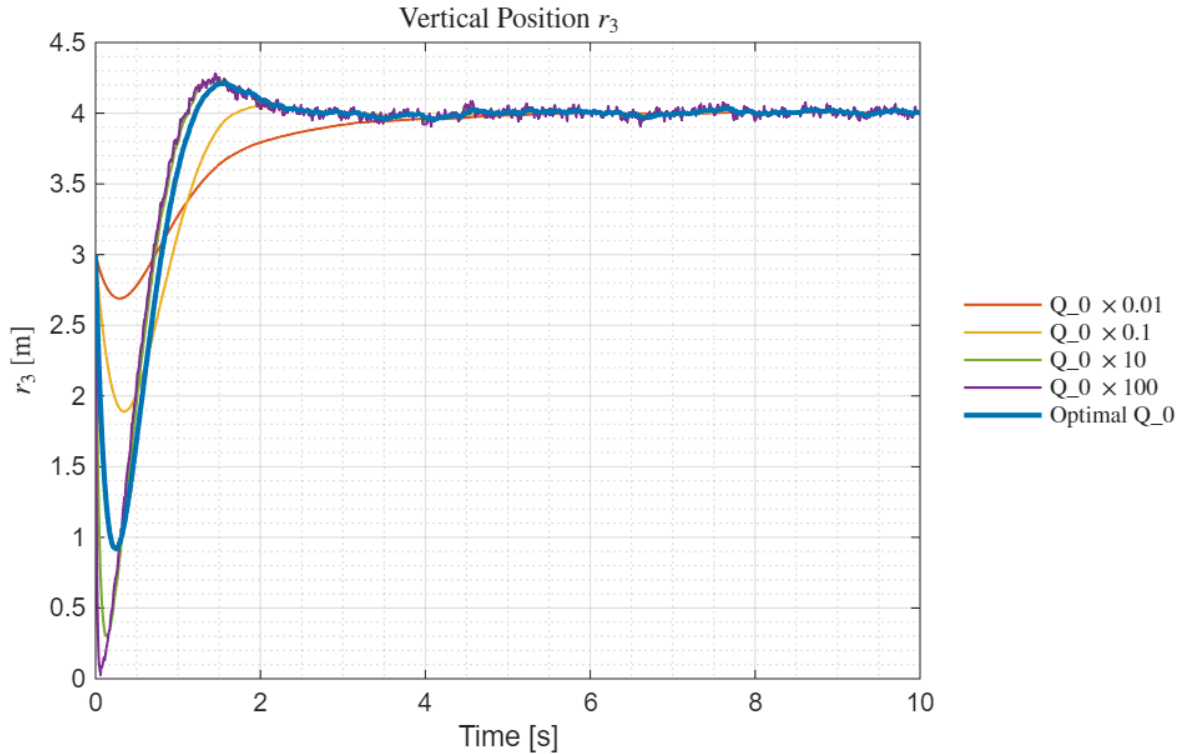


Figure 2: Q6 — Q_0 sensitivity analysis: vertical position r_3 .

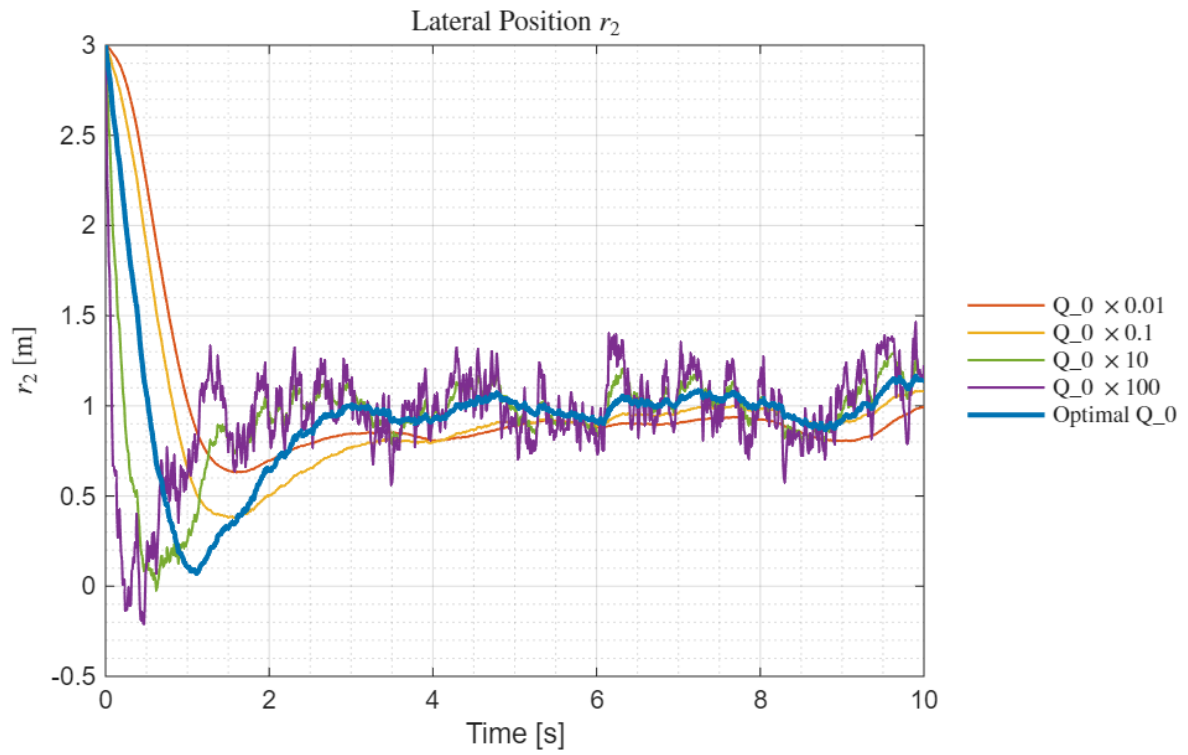


Figure 3: $Q_6 - Q_0$ sensitivity analysis: lateral position r_2 .

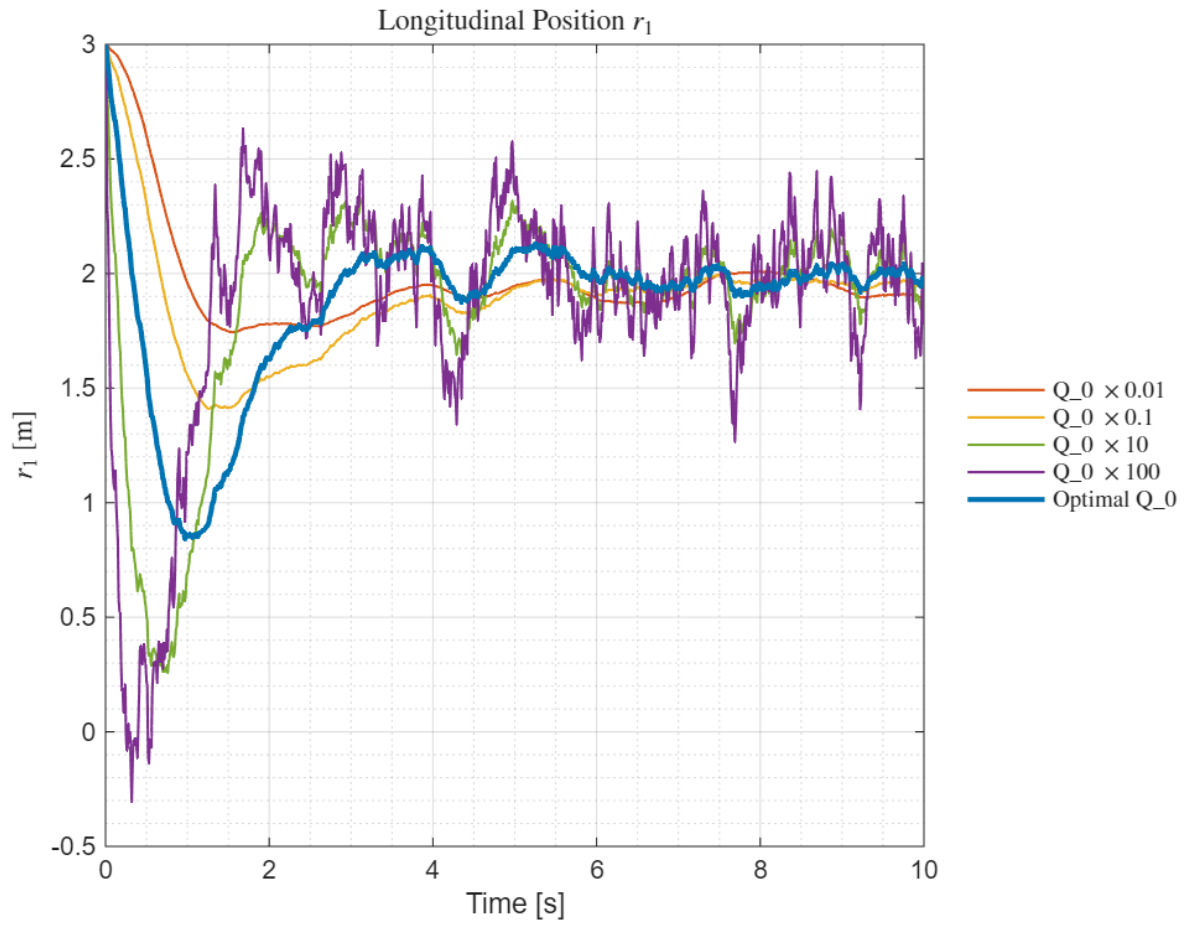


Figure 4: Q6 — Q_0 sensitivity analysis: longitudinal position r_1 .

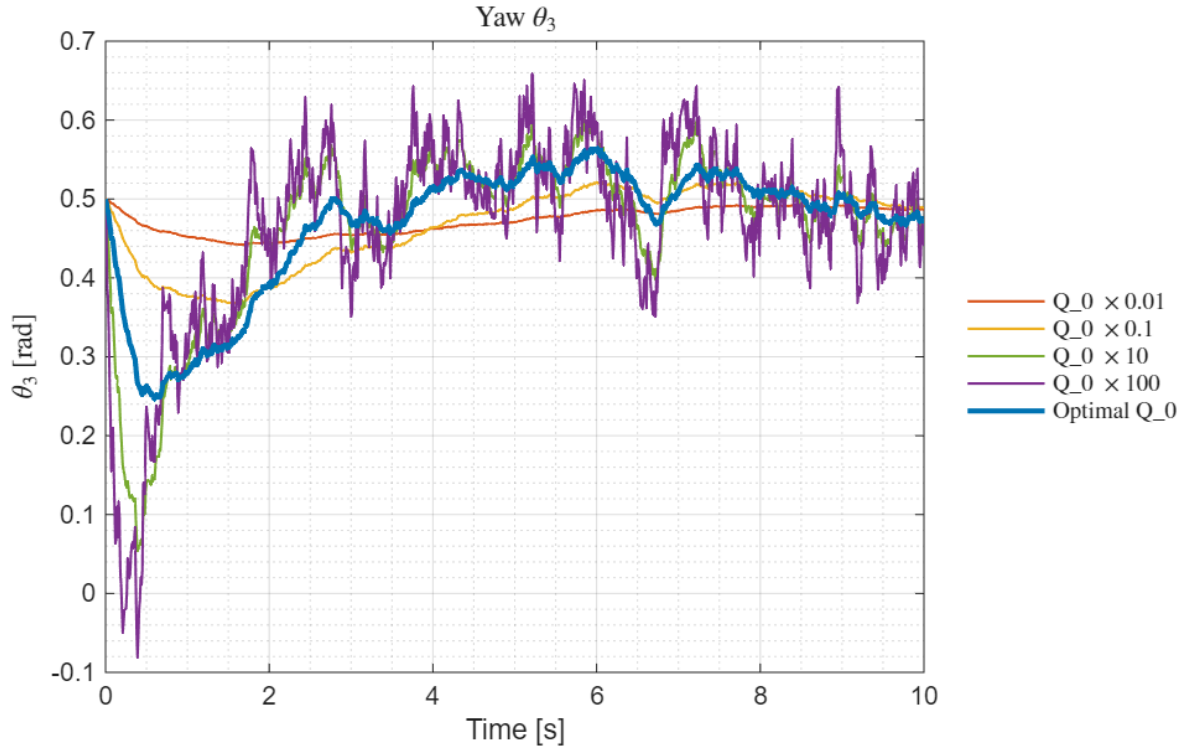


Figure 5: Q6 — Q_0 sensitivity analysis: yaw angle θ_3 .

The vertical position r_3 shows little sensitivity to Q_0 changes compared to the other channels. Its low measurement noise variance ($\sigma_{M,r_3}^2 = 1.80 \times 10^{-3} \text{ m}^2$) makes the measurements inherently reliable, so the filter performs well regardless of the assumed process noise level. Only the most extreme case ($Q_0 \times 100$) shows a slightly noisier steady-state estimate. For r_2 , r_1 , and θ_3 , the effect is clear:

- **High Q_0 ($\times 10$, $\times 100$)** — the filter assumes the model is unreliable and trusts the measurements more. The estimate becomes noisier but tracks fast changes more aggressively.
- **Low Q_0 ($\times 0.1$, $\times 0.01$)** — the filter assumes the model is highly accurate and relies on it more. The estimate is smoother but may lag behind sudden changes in the true state. Physically, this means the filter is overconfident in the model — unmodelled disturbances and linearization errors are not accounted for, which can lead to persistent estimation errors.
- **Optimal Q_0** — provides the best balance between noise rejection and tracking speed, consistent with the minimum variance optimality of the Kalman-Bucy filter.

1.7 Sensitivity Analysis of R_0

The measurement noise covariance matrix R_0 represents the confidence level in the sensor measurements. The same five scaling factors were tested while keeping Q_0 fixed at its optimal value. The results are shown in Figures 6–9.

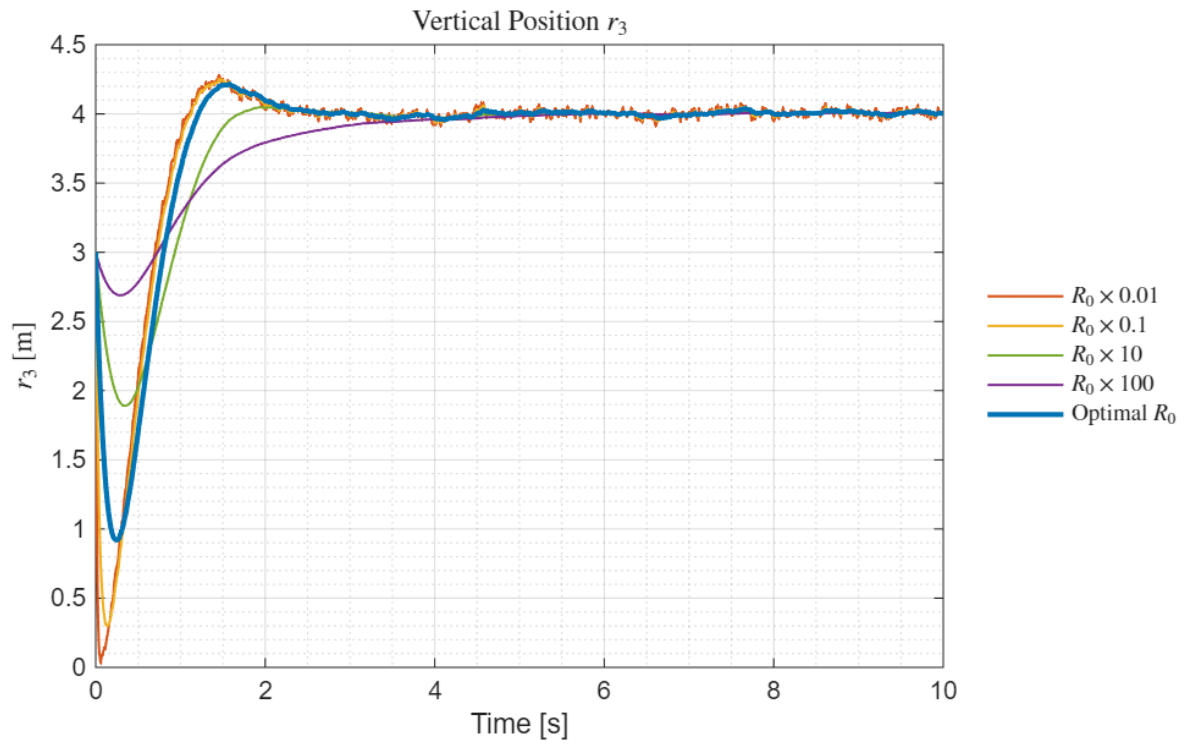


Figure 6: Q7 — R_0 sensitivity analysis: vertical position r_3 .

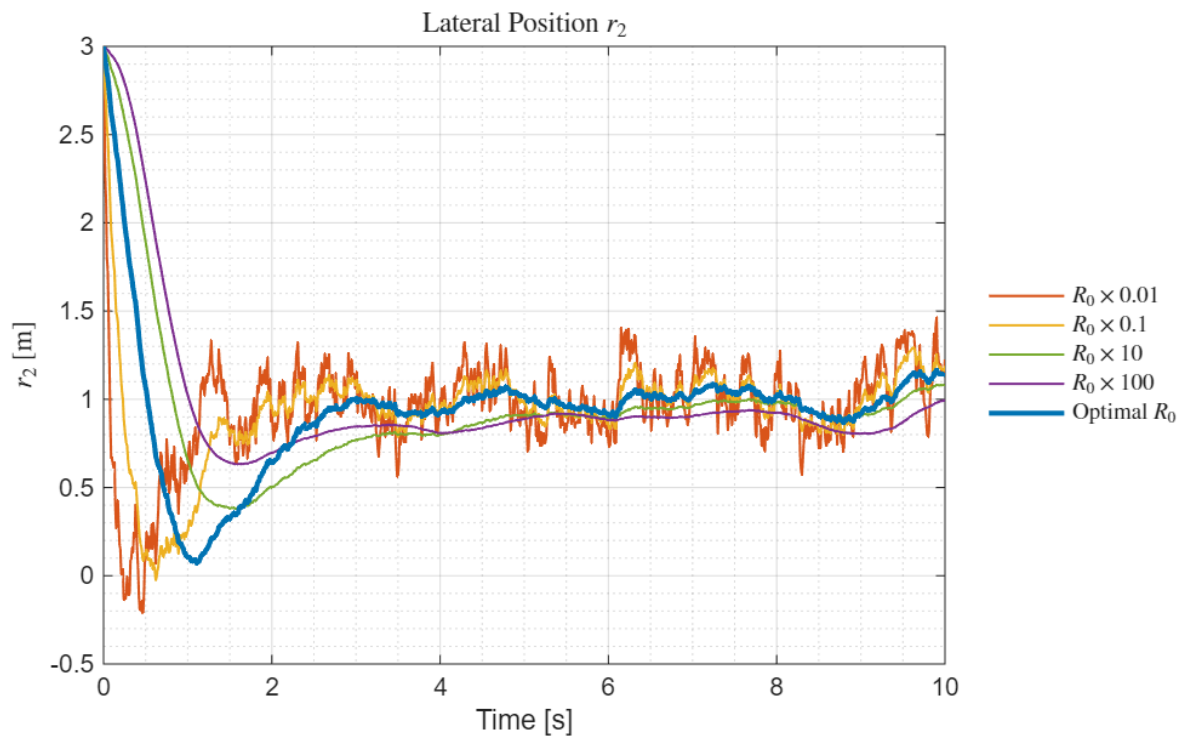


Figure 7: Q7 — R_0 sensitivity analysis: lateral position r_2 .

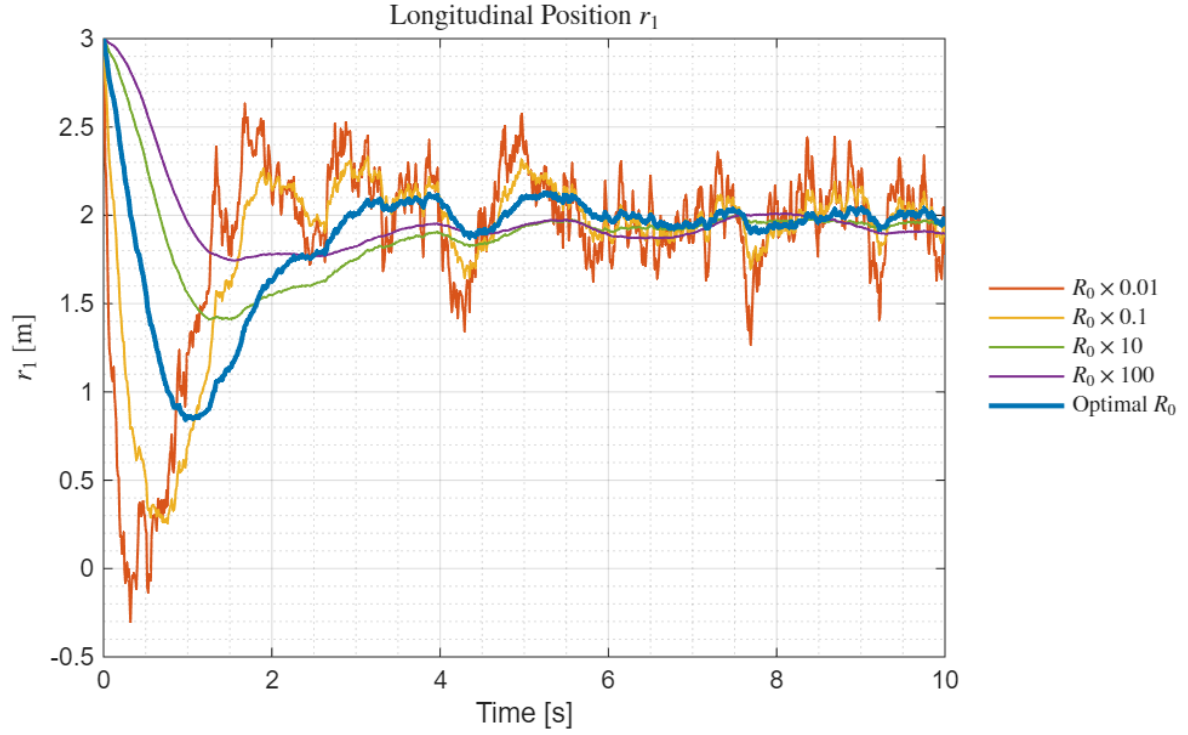


Figure 8: Q7 — R_0 sensitivity analysis: longitudinal position r_1 .

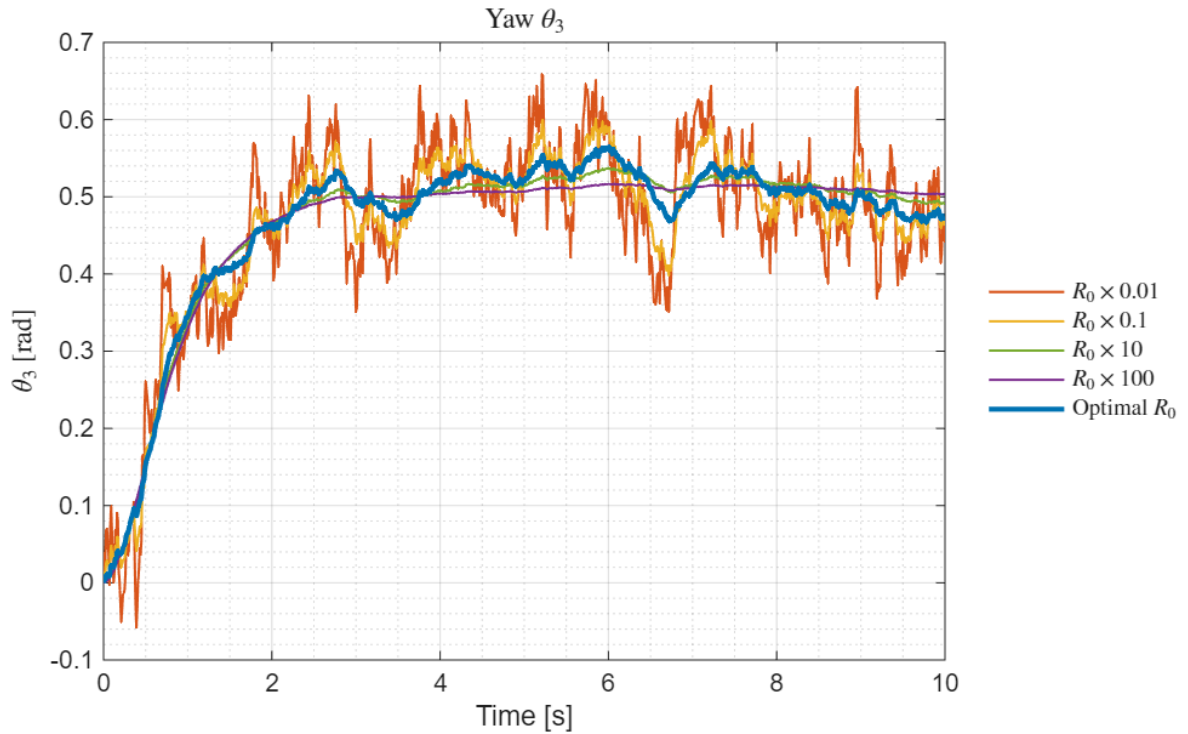


Figure 9: Q7 — R_0 sensitivity analysis: yaw angle θ_3 .

The effect of R_0 is the dual of Q_0 . The vertical position r_3 again shows little sensitivity across all tested scales for the same reasons as in Q6.

For r_2 , r_1 , and θ_3 :

- **Low** R_0 ($\times 0.1$, $\times 0.01$) — the filter assumes sensors are very accurate and follows measurements closely. The

estimate becomes noisier — the same effect as a high Q_0 .

- **High** R_0 ($\times 10$, $\times 100$) — the filter assumes sensors are unreliable and relies more on the model. The estimate is smoother but lagged — the same effect as a low Q_0 .
- **Optimal** R_0 — again provides the best balance.

This confirms that what governs the estimator behaviour is the **ratio** Q_0/R_0 , not the absolute values individually. Increasing Q_0 by a factor of 100 is qualitatively equivalent to decreasing R_0 by the same factor. The filter is robust to small perturbations but degrades noticeably when the assumed statistics deviate by two or more orders of magnitude from the true values.

1.8 Kalman-Bucy Observer on the Nonlinear Model

1.8.1 Observer Convergence

By including process noise and measurement noise in the nonlinear model it becomes clear that the observer remains effective for a more realistic system, as shown in Figure 10.

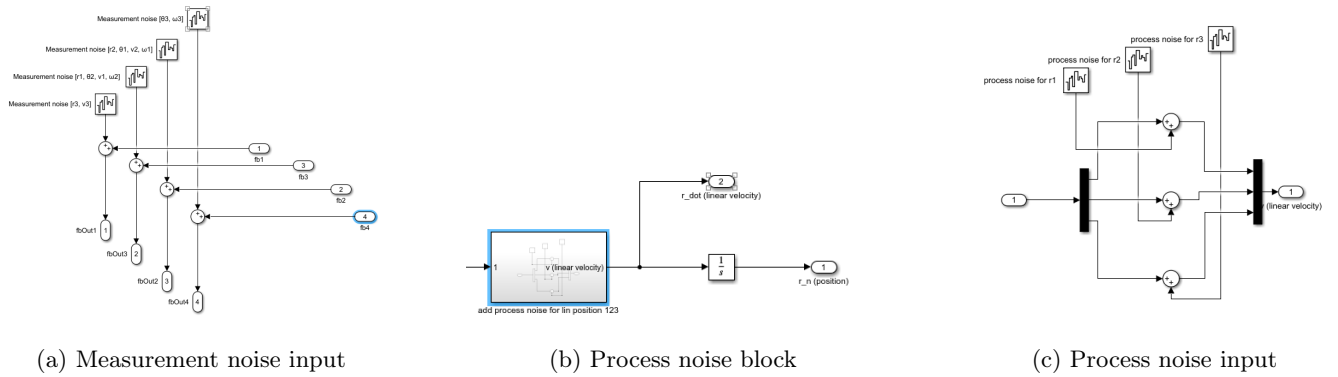


Figure 10: Noise blocks added to the nonlinear model.

To deal with the noise a Kalman-Bucy observer is included, as shown in Figure 11. The observer has the optimal L values computed through the algebraic Riccati equation.

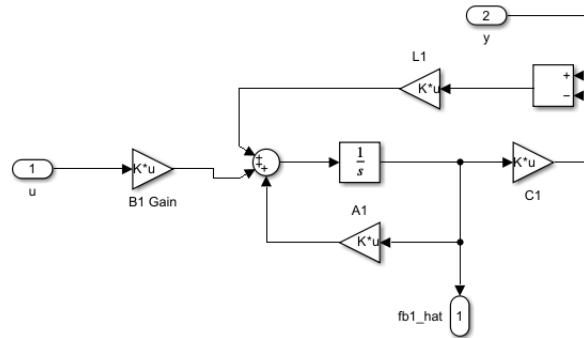


Figure 11: Kalman-Bucy observer block in the nonlinear model.

Generally estimation is less smooth than in the linear case because of the nonlinearities but it can be stated that the observer works well with the nonlinear model as shown in Figures 12–15.

The x Cartesian position of the drone starts with the initial condition of $x = 0$ but the observer's initial estimate is $\hat{x} = 3$ m. The reference goal for the regulator is $x^{\text{ref}} = 4$ m, as shown in Figure 12.

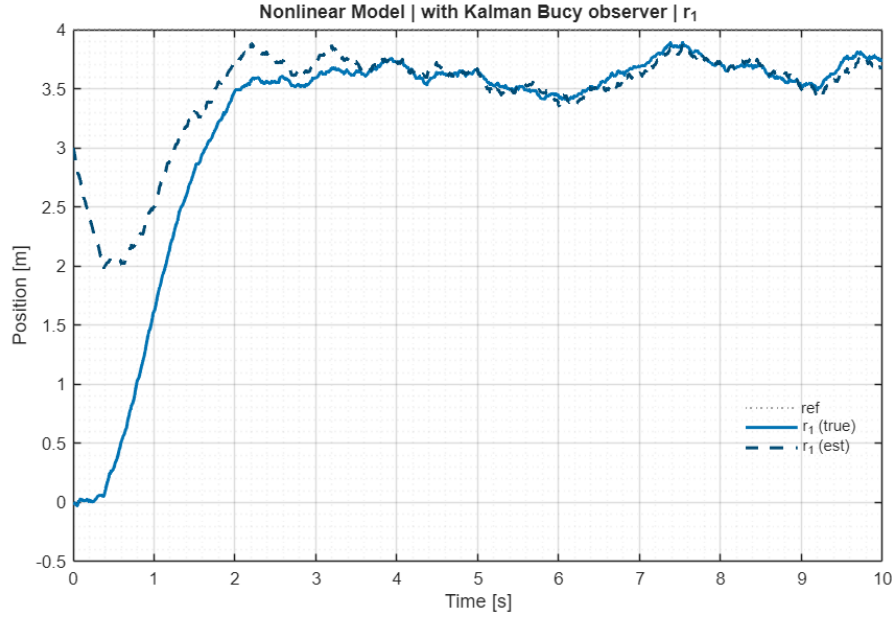


Figure 12: r_1 response and estimated response $\hat{r}_1$.

The y Cartesian position of the drone starts with the initial condition of $y = 0$ but the observer's initial estimate is $\hat{y} = 3$ m. The reference goal for the regulator is $y^{\text{ref}} = 4$ m, as shown in Figure 13.

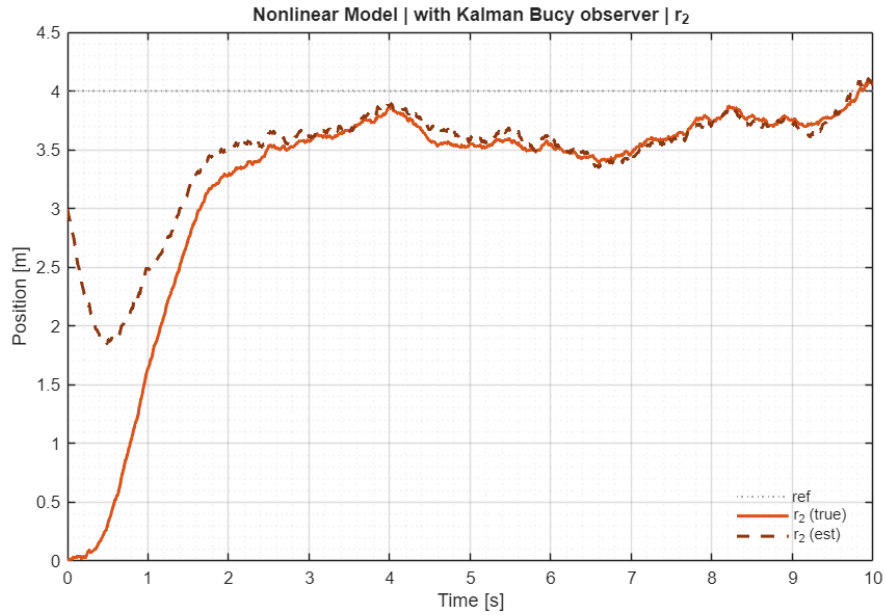


Figure 13: r_2 response and estimated response $\hat{r}_2$.

The z Cartesian position of the drone starts with the initial condition of $z = 0$ but the observer's initial estimate is $\hat{z} = 3$ m. The reference goal for the regulator is $z^{\text{ref}} = 4$ m, as shown in Figure 14.

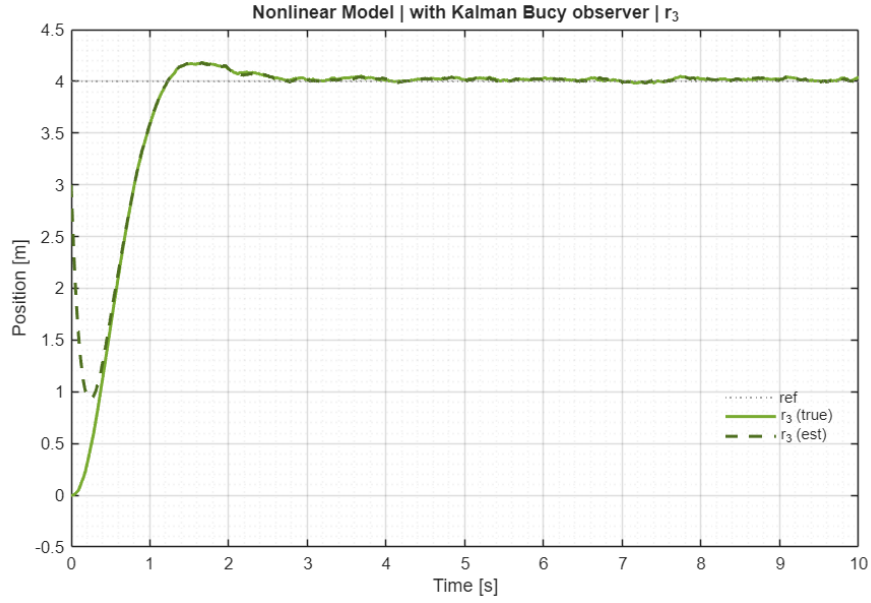


Figure 14: r_3 response and estimated response $\hat{r}_3$.

The yaw angle θ_3 of the drone starts with the initial condition of $\theta_3 = 0$ but the observer's initial estimate is $\hat{\theta}_3 = 0.5$ rad. The reference goal for the regulator is $\theta_3^{\text{ref}} = 1$ rad. The actual yaw dynamics are highly variable, featuring pronounced disturbances and rapid changes. The observer successfully smooths these fluctuations to provide a stable estimate, as shown in Figure 15. This filtering inherently reduces the system's responsiveness to sudden yaw maneuvers.

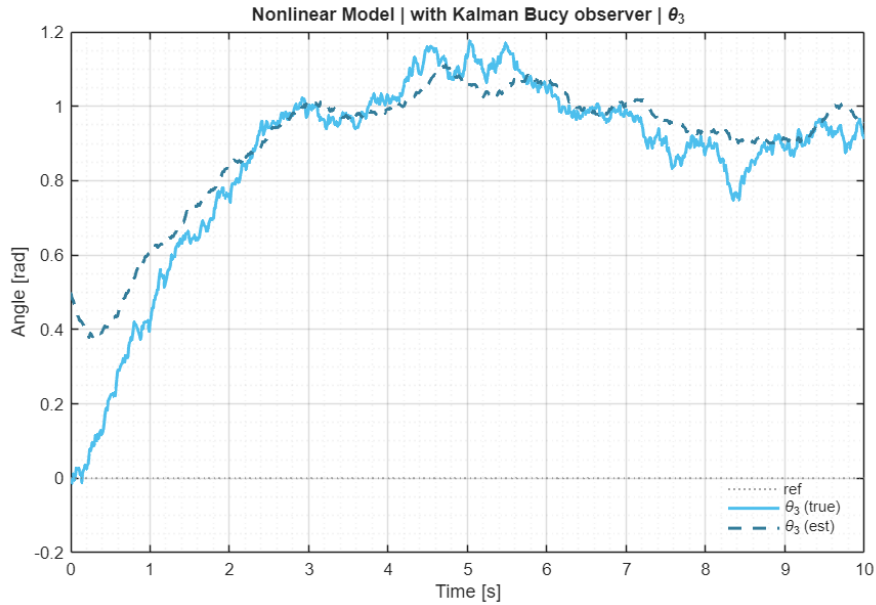


Figure 15: θ_3 response and estimated response $\hat{\theta}_3$.

In all these cases it can be stated that the observer quickly corrects its wrong initial estimate. For the rest of the simulation time the estimation is in good agreement with the actual trajectory of the drone. The variation in the correction speed is explained by the different amount of noise that is present in each state which produces different L values in the observer.

1.8.2 Sensitivity Analysis of Q_0

As observed with the linear model, we can evaluate the Kalman-Bucy observer's behavior under varying Q_0 parameters. The results are shown in Figures 16–19.

Modifying Q_0 within the nonlinear framework emphasizes the inherent compromise between tracking speed and estimation stability. Assigning a low value to Q_0 implies an underestimation of system disturbances, which introduces noticeable tracking lag alongside minor overshoots. Conversely, an excessively high Q_0 exaggerates the model's uncertainty, resulting in highly oscillatory estimates.

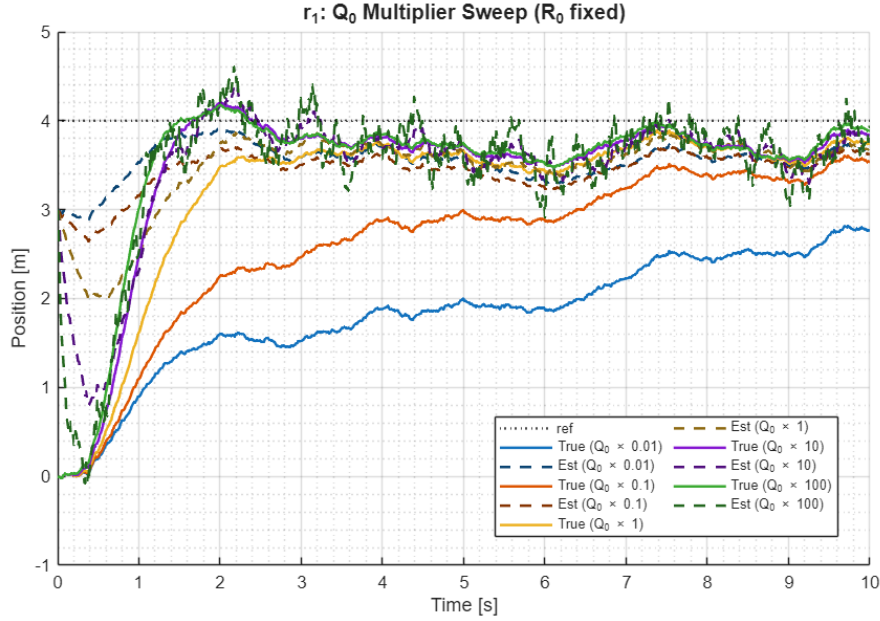


Figure 16: r_1 response for varying Q_0 values.

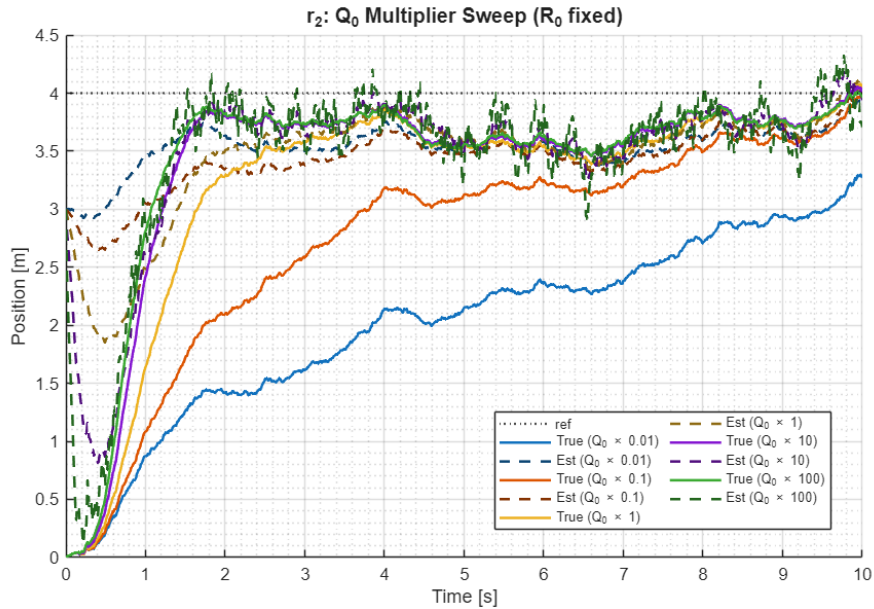


Figure 17: r_2 response for varying Q_0 values.

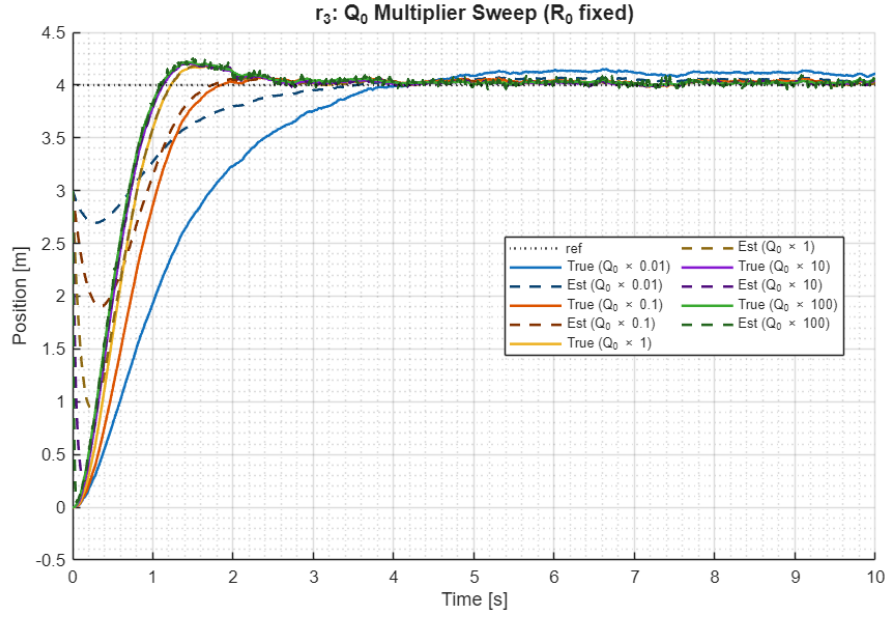


Figure 18: r_3 response for varying Q_0 values.

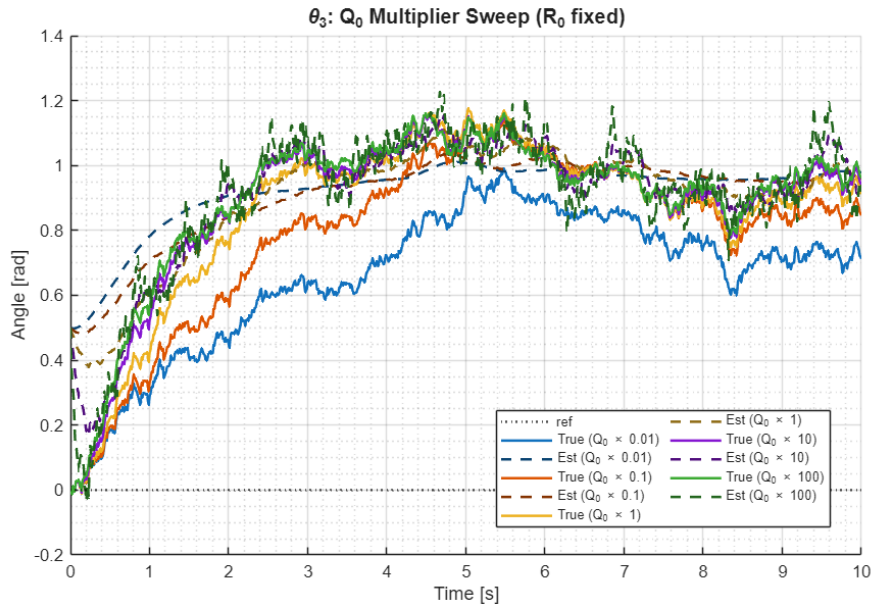


Figure 19: θ_3 response for varying Q_0 values.

Some very low multipliers cause unacceptable lag in the drone's response time and the highest multiplier value has estimator oscillations many times larger than the actual oscillations of the states. Ultimately, intermediate values provide the optimal balance, confirming that foundational tuning strategies remain applicable regardless of the system's underlying complexity.

1.8.3 Sensitivity Analysis of R_0

As observed with the linear model, we can evaluate the Kalman-Bucy observer's behavior under varying R_0 parameters. The results are shown in Figures 20–23.

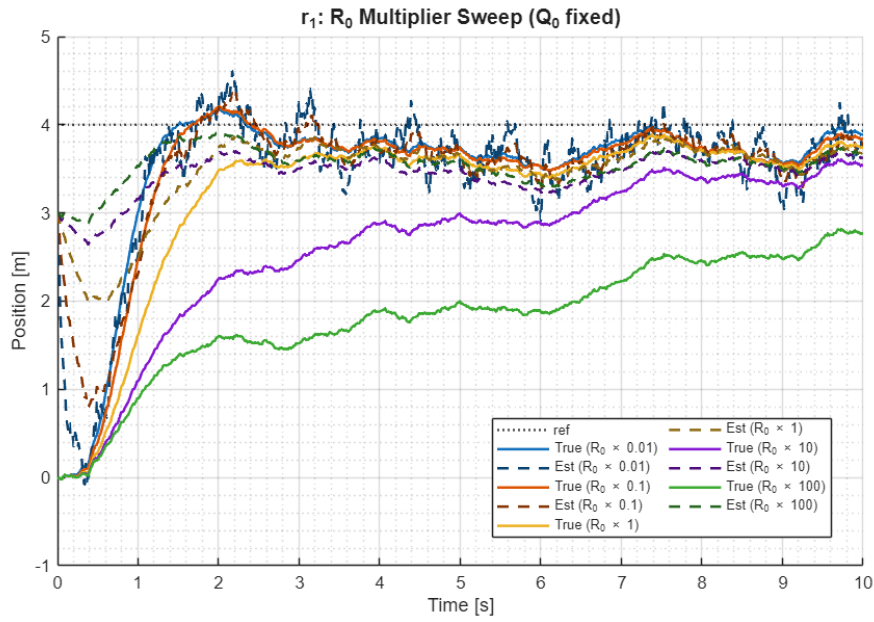


Figure 20: r_1 response for varying R_0 values.

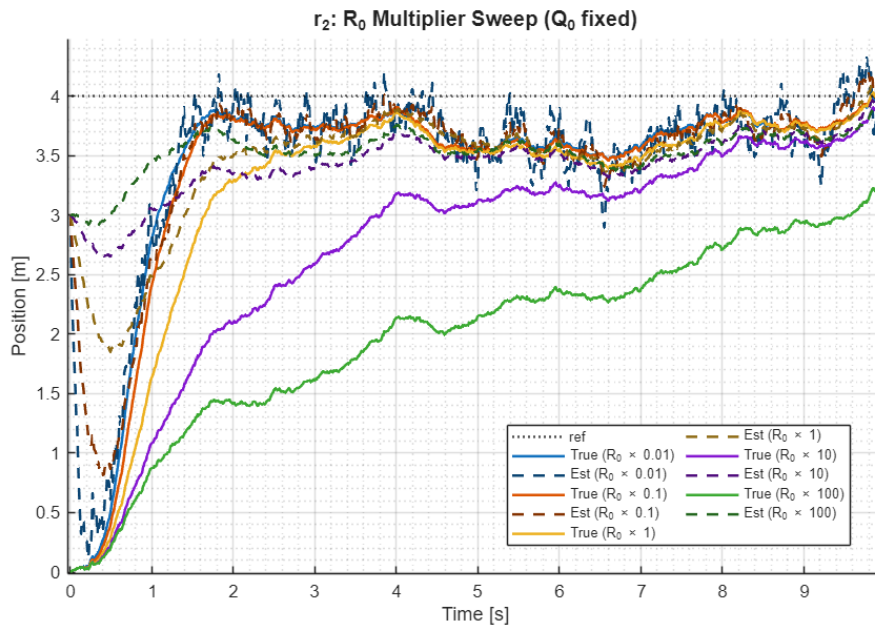


Figure 21: r_2 response for varying R_0 values.

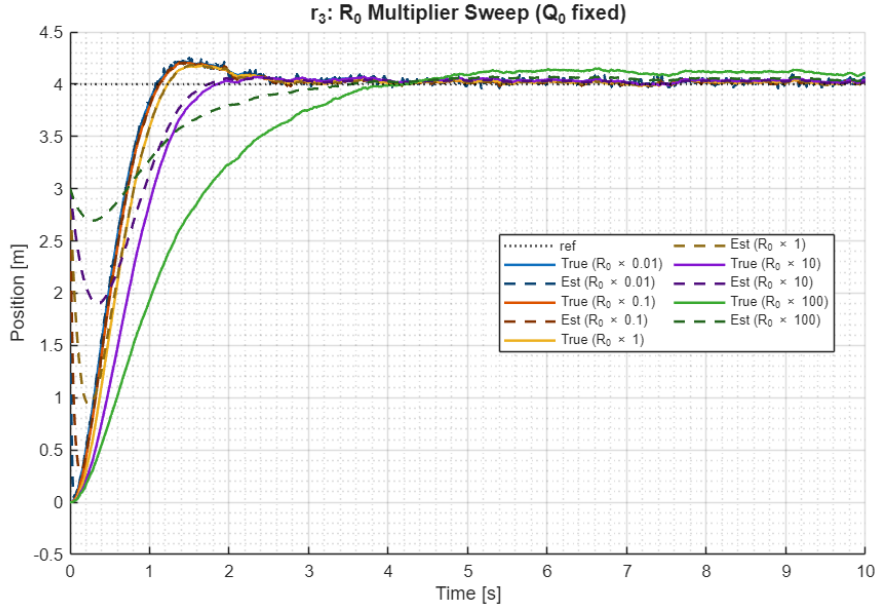


Figure 22: r_3 response for varying R_0 values.

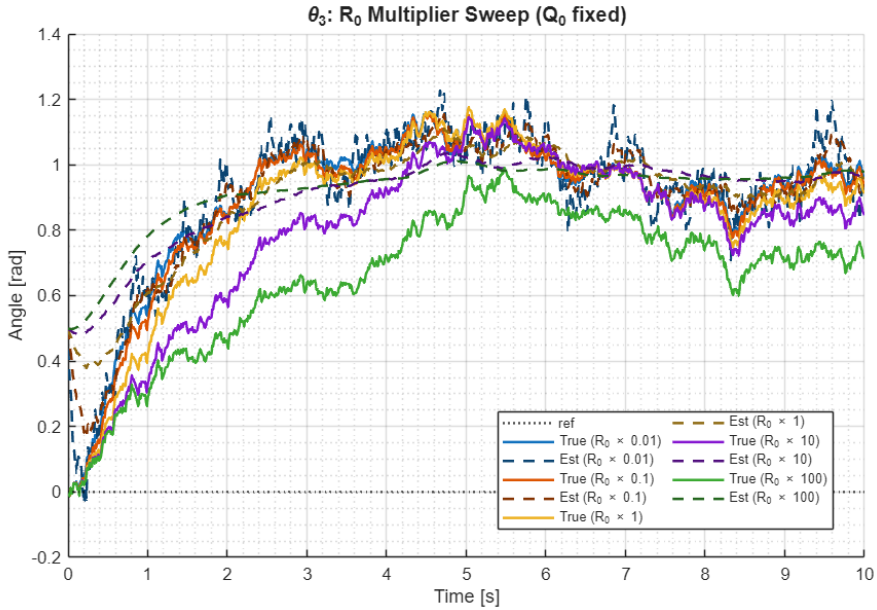


Figure 23: θ_3 response for varying R_0 values.

A low R_0 leads to noisy estimates that are more sensitive to sensor data, while a high R_0 results in a smoother but slower estimate. An intermediate setting effectively filters noise without sacrificing accuracy, ensuring reliable performance in conditions closer to real-world operation. The behavior of the change in responses when changing R_0 is almost exactly the opposite of Q_0 .

Compared to the linearized model, the nonlinear system introduces additional coupling between states and unmodelled dynamics, which manifest as slightly larger estimation errors during transients. Nevertheless, the observer successfully corrects its initial estimate and tracks the true trajectory in all channels, confirming that the Kalman-Bucy design remains effective beyond the linearization assumptions.

2 LQR Servo Controller

2.1 LQR Servo Controller Formulation

To construct the servo controller, the drone system must be formulated in an augmented state-space representation. The augmented state vector for each subsystem includes an additional integral state $\xi_i = \int e_i dt$, where e_i is the tracking error of the controlled output, such that:

$$\bar{x}_i = \begin{bmatrix} x_i \\ \xi_i \end{bmatrix} \quad (7)$$

This integral action eliminates steady-state error in the presence of constant disturbances. The resulting augmented system is defined as follows:

For the vertical displacement subsystem:

$$\bar{A}_1 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}, \quad \bar{B}_1 = \begin{bmatrix} 0 \\ \frac{1}{m_n} \\ 0 \end{bmatrix}, \quad \bar{C}_1 = [1 \quad 0 \quad 0] \quad (8)$$

For the lateral displacement subsystem:

$$\bar{A}_2 = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & g & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \bar{B}_2 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{J_{11}} \\ 0 \end{bmatrix}, \quad \bar{C}_2 = [1 \quad 0 \quad 0 \quad 0 \quad 0] \quad (9)$$

For the longitudinal displacement subsystem:

$$\bar{A}_3 = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & -g & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \bar{B}_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{J_{22}} \\ 0 \end{bmatrix}, \quad \bar{C}_3 = [1 \quad 0 \quad 0 \quad 0 \quad 0] \quad (10)$$

For the yaw rotation subsystem:

$$\bar{A}_4 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}, \quad \bar{B}_4 = \begin{bmatrix} 0 \\ \frac{1}{J_{33}} \\ 0 \end{bmatrix}, \quad \bar{C}_4 = [1 \quad 0 \quad 0] \quad (11)$$

The LQR gain $\bar{K}_i = [K_i \quad k_{i,\text{int}}]$ for each subsystem is computed by solving the algebraic Riccati equation, minimising the quadratic cost function:

$$J = \int_0^\infty (\bar{x}_i^\top Q_i \bar{x}_i + u_i^\top R_i u_i) dt \quad (12)$$

The resulting control law for each subsystem is:

$$u_i = -\bar{K}_i \bar{x}_i \quad (13)$$

The perturbation input vector is:

$$\mathbf{u} = [F'_3 \quad M'_1 \quad M'_2 \quad M'_3]^\top \quad (14)$$

Utilizing the same position reference values employed for observer validation (ref=4), the LQR demonstrated a robust ability to guide the system to the desired state and maintain stability. This performance was verified in a simulation featuring environmental noise, as shown in Figure 24.

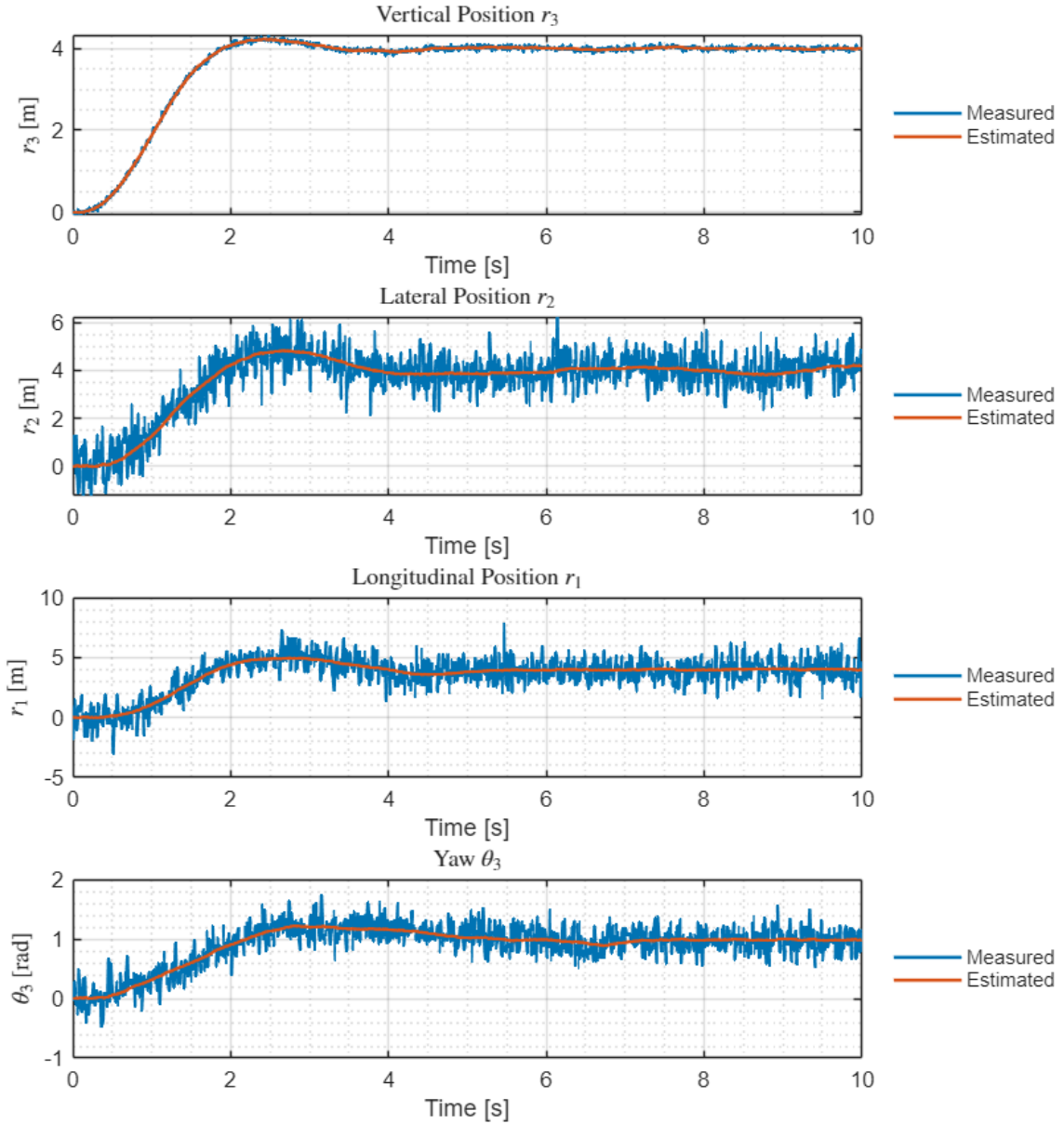


Figure 24: Response of the linear model with the LQR servo controller.

2.2 Role and Tuning of the Q and R Matrices

The LQR servo controller is obtained by minimizing the quadratic cost function

$$J = \int_0^\infty (\bar{x}_i^\top Q_i \bar{x}_i + u_i^\top R_i u_i) dt, \quad (15)$$

where $Q_i \succeq 0$ and $R_i \succ 0$ are weighting matrices.

The matrix Q_i weights the augmented state vector $\bar{x}_i$, which includes the states of the subsystem together with the integrator state introduced in the servo formulation. This means that Q_i determines how much the controller penalizes deviations of the states from the desired behaviour. If the entries of Q_i are increased, the controller gives more importance to reducing those state errors, which usually leads to a faster and more aggressive response.

The matrix R_i weights the control input u_i . Its role is to penalize large control actions. Therefore, increasing R_i makes the controller less aggressive, producing smoother actuation and reducing the control effort, although this usually results in a slower response.

For each subsystem in this project, the input is scalar, so

$$R_i \in R^{1 \times 1}. \quad (16)$$

The matrix Q_i must have the same dimension as the augmented state vector $\bar{x}_i$. Since the servo controller adds one integrator state to the original subsystem, if a subsystem has n_i states then the augmented system has $n_i + 1$ states, and therefore

$$Q_i \in R^{(n_i+1) \times (n_i+1)}. \quad (17)$$

In this case,

$$Q_1, Q_4 \in R^{3 \times 3}, \quad Q_2, Q_3 \in R^{5 \times 5}, \quad (18)$$

because subsystems 1 and 4 have two states, while subsystems 2 and 3 have four states.

A practical way to choose the initial values of Q_i and R_i is to use Bryson's rule,

$$Q_{i,jj} = \frac{1}{x_{i,j,\max}^2}, \quad R_i = \frac{1}{u_{i,\max}^2}, \quad (19)$$

which scales the states and the input according to their maximum acceptable values.

After that, the matrices are tuned iteratively depending on the desired behaviour. In general, increasing Q_i relative to R_i makes the controller track the reference more quickly, but also increases the control effort. On the other hand, increasing R_i relative to Q_i gives smoother control signals and lower actuation effort, but the response becomes slower. Therefore, tuning Q_i and R_i is essentially a compromise between fast tracking and moderate control effort.

2.3 Influence of Q and R on the Vertical Subsystem

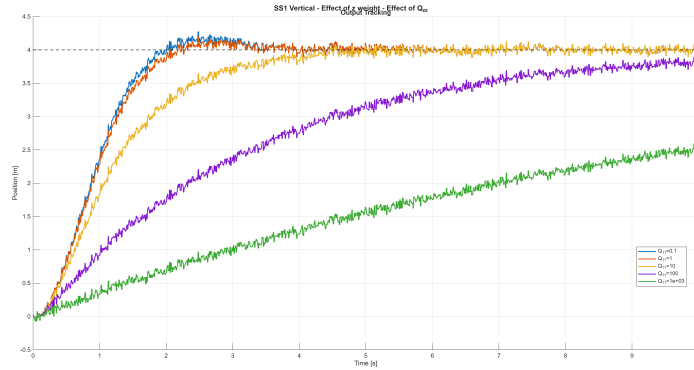


Figure 25: Response for tuning of parameter Q_{11} for z

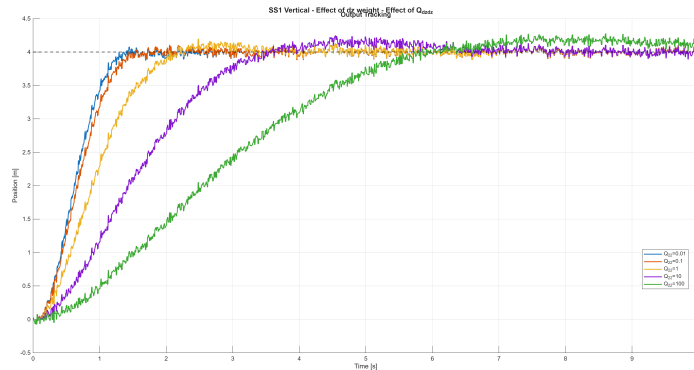


Figure 26: Response for tuning of parameter Q_{22} for $\dot{z}$

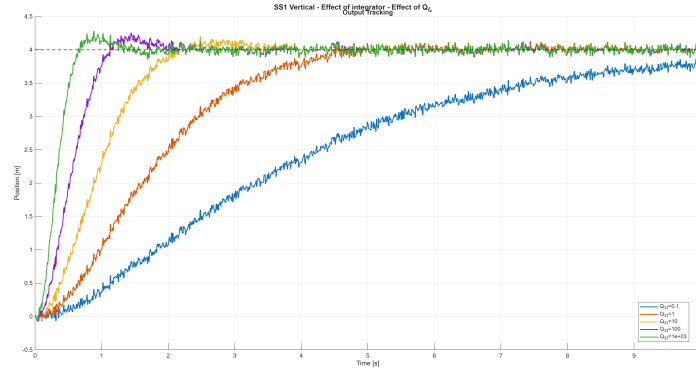


Figure 27: Response for tuning of parameter Q_{33} for integrator

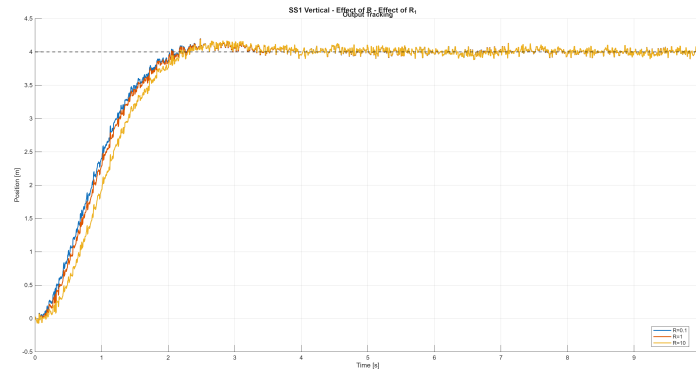


Figure 28: Response for tuning of parameter R

The influence of the input penalty R and each diagonal of $Q = \text{diag}(Q_{11}, Q_{22}, Q_{33})$ was evaluated the time response. Figure 28 shows the effect of varying R . As expected, increasing R makes the control action less aggressive. The response becomes slower, with a larger rise time, while the final tracking accuracy remains approximately unchanged. Lower values of R produce a faster approach to the reference, at the cost of a slightly more aggressive transient.

Figure 25 shows the effect of the weight Q_{11} , associated with the state z . In this case, the simulations show that larger values of Q_{11} do not improve tracking speed; on the contrary, they lead to a much slower approach to the 4m reference, used for these simulation. The best transient is obtained for the smaller values $Q_{11} = 0.1$ and $Q_{11} = 1$, while very large values such as $Q_{11} = 100$ and $Q_{11} = 1000$ make the response unusable. Therefore, for this servo LQR, excessively penalizing the altitude state reduces the effectiveness of the reference tracking.

Figure 26 shows the effect of the velocity weight Q_{22} , associated with $\dot{z}$. Increasing Q_{22} also slows down the transient response. For small values ($Q_{22} = 0.01$ or 0.1), the altitude reaches the reference quickly, while larger values produce a more slower behaviour. Although $\dot{z}$ weighting is usually related to damping, in these simulations excessive penalization of vertical velocity mainly reduces the response speed rather than improving the overall transient.

Finally, Figure 27 shows the effect of the integrator weight Q_{33} , associated with ξ_z . This is the parameter with the clearest positive impact on tracking performance. Increasing Q_{33} substantially accelerates the convergence to the reference and improves steady-state error rejection. For low values such as $Q_{33} = 0.1$, the system converges very slow and does not reach the reference within the simulation time. For higher values ($Q_{33} = 10, 100$, and 1000), the output reaches 4m much faster, showing the importance of the integral block in servo tracking.

Overall, the results show that, for the vertical subsystem, the most effective way to improve tracking is to increase the weight on the integrator state, while large values of Q_{11} , Q_{22} , or R make the response slow.

2.4 LQR Servo Controller for the Longitudinal Subsystem

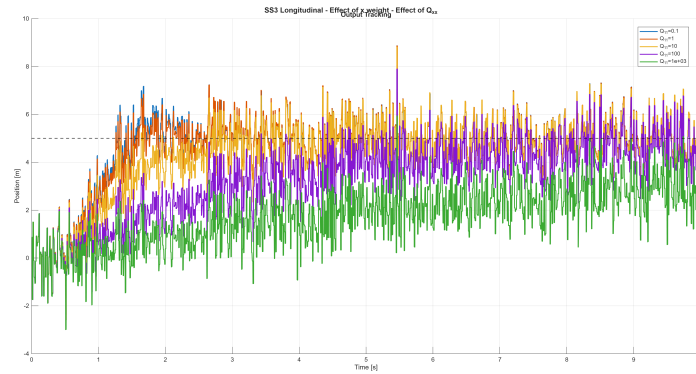


Figure 29: Response for tuning parameter Q_{11} for x

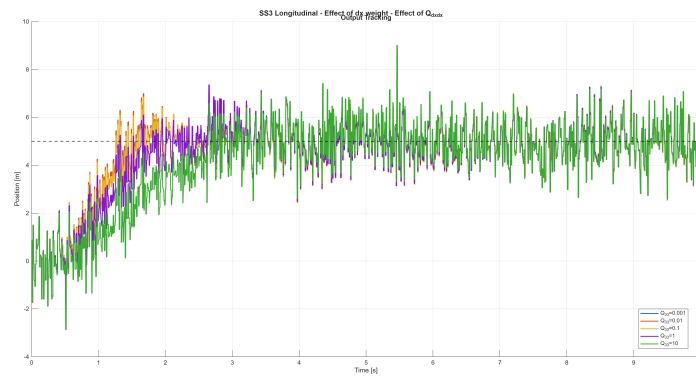


Figure 30: Response for tuning parameter Q_{33} for $\dot{x}$

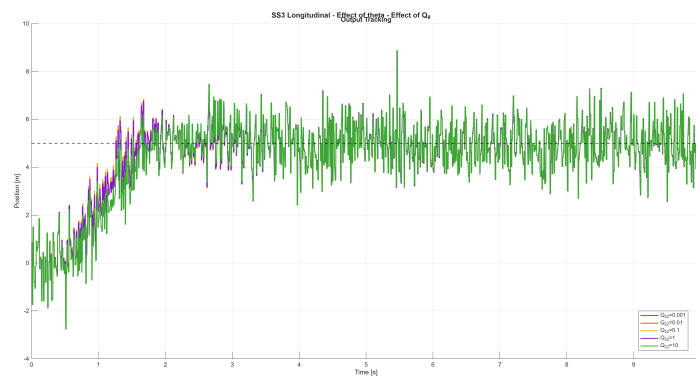


Figure 31: Response for tuning parameter Q_{22} for θ

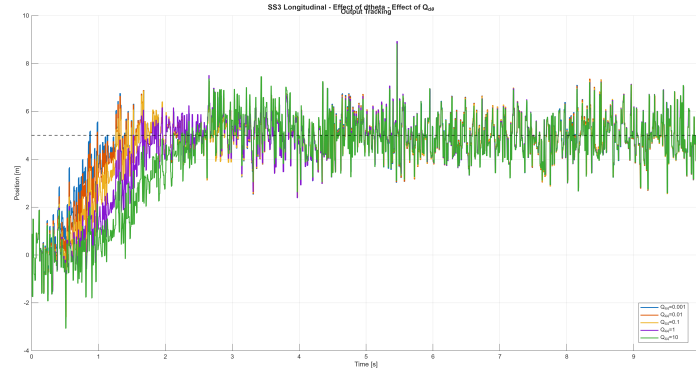


Figure 32: Response for tuning parameter Q_{44} for $\dot{\theta}$

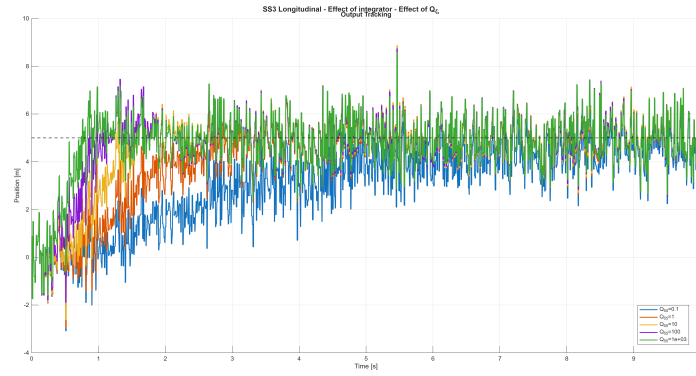


Figure 33: Response for tuning parameter Q_{55} for integrator

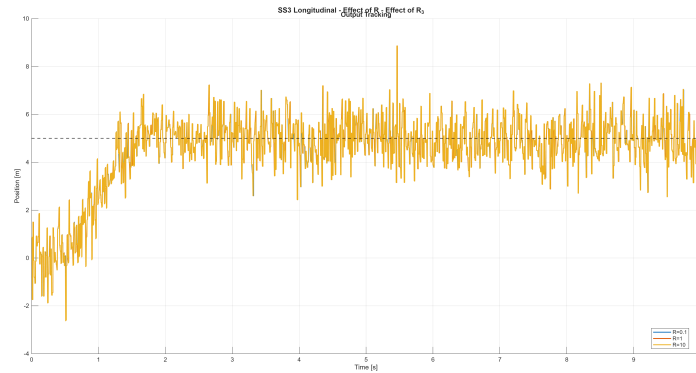


Figure 34: Response for tuning parameter R

The influence of the input weight R and of each diagonal entry of the matrix

$$Q = \text{diag}(Q_{11}, Q_{22}, Q_{33}, Q_{44}, Q_{55})$$

was evaluated from the time response.

Figure 34 shows the effect of varying R . There is no change on the response for different gains

Figure 29 shows the effect of the position weight Q_{11} , associated with the state x . For small and moderate values ($Q_{11} = 0.1, 1, 10$), the output is able to approach the reference. But, for larger values ($Q_{11} = 100, 1000$), the tracking performance becomes significantly worse, with a much slower response and larger deviation from the reference. So just like before excessively increasing the weight on x does not improve tracking and instead degrades the closed-loop response.

Figure 31 shows the effect of the weight on θ , namely Q_{22} . The responses for different values are quite similar, which indicates that this weight has only a limited influence on the measured longitudinal position in the tested range. Only

for the largest value does the response become slightly more irregular. This suggests that the pitch-angle mainly affects the internal attitude dynamics, while its influence on the output tracking is less pronounced.

Figure 30 and Figure 32 show the effect of the weights on $\dot{x}$ and $\dot{\theta}$, respectively. In both cases, the influence on the measured output is relatively small for low and moderate values, while very large weights tend to make the response a little slower. But these weights do not produce a significant improvement in the tracking of x under the tested conditions, although they may still help limit excessive velocity and pitch-rate variations internally.

Finally, Figure 33 shows the effect of the integrator weight Q_{55} , associated with ξ_x . This is the parameter with the clearest positive effect on the longitudinal tracking performance. Increasing Q_{55} makes the output approach the reference more quickly and improves steady-state error rejection. For small values, the system converges slowly and remains below the reference for a big portion of the simulation, however the larger values makes the controller much more effective in driving the position toward the desired value.

Overall, the longitudinal subsystem is strongly affected by noise, and the clearest improvement in tracking is obtained by increasing the integrator weight. On the other hand, excessively large state weights do not improve the response and may even degrade it.

2.5 Performance with State Observer in the Loop

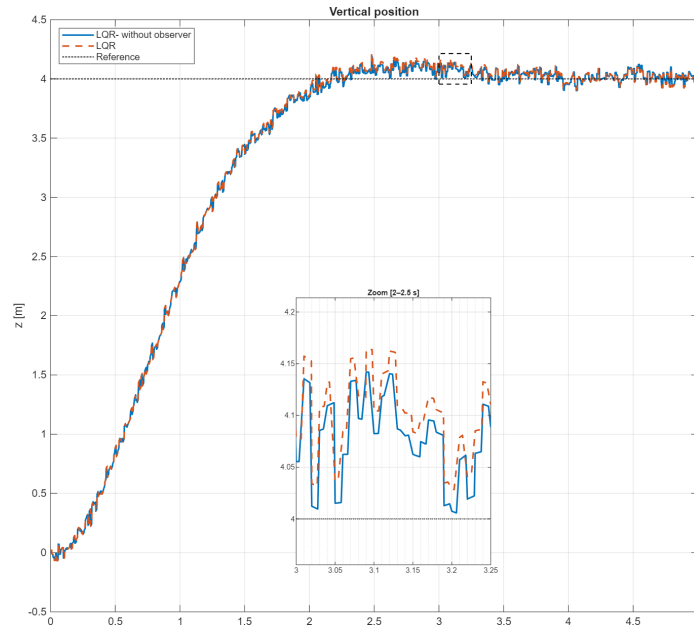


Figure 35: Response with and without observer for vertical subsystem

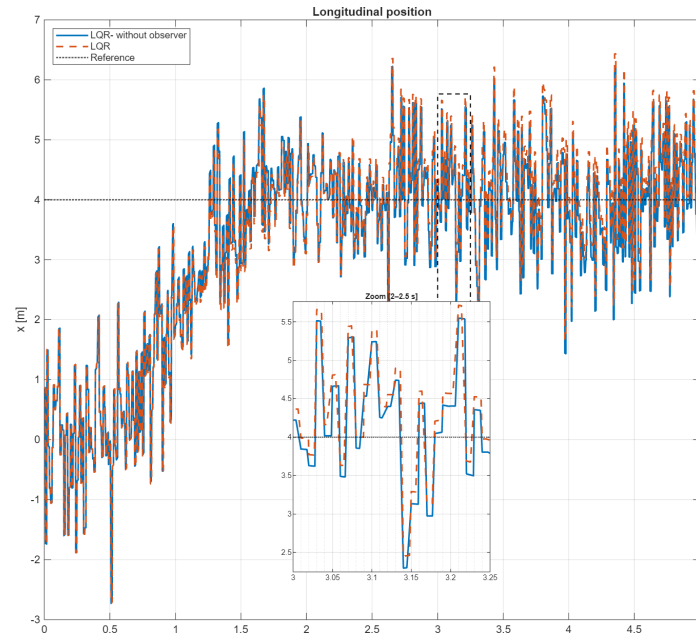


Figure 36: Response with and without observer for longitudinal subsystem

Including the state observer in the loop does not significantly change the controller performance. From the plots in Question 13, the responses with and without the observer are almost superimposed, both in the vertical and longitudinal motions. This indicates that the rise time, settling time, and steady-state tracking accuracy remain essentially the same. A small difference can be observed in the zoomed regions, where the response with the observer presents slightly different oscillation amplitudes around the reference. However, this effect is minor and does not noticeably affect the overall closed-loop behavior.

This is also consistent with the expected theoretical result: when the observer is properly designed and converges sufficiently fast, the estimated states become very close to the real states, so the controller behaves almost like a full-state feedback controller. Therefore, only small differences due to estimation dynamics, measurement noise, or model mismatch are expected.

2.6 LQR Servo Controller on the Nonlinear Model

2.6.1 Vertical Subsystem: Individual Reference r_3

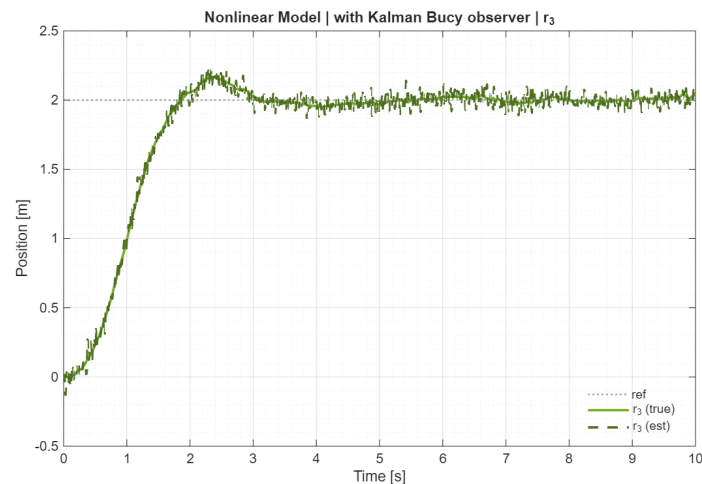


Figure 37: Response pre-tuning for altitude

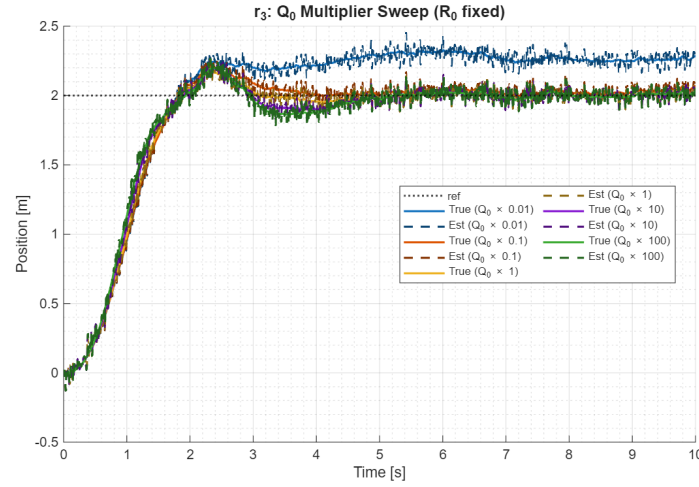


Figure 38: Response tuning for altitude for Q_0

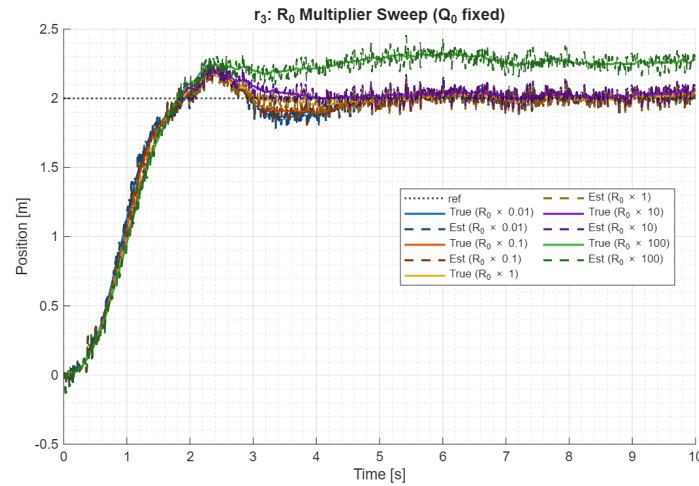


Figure 39: Response tuning for altitude for R_0

2.7 Vertical Subsystem – Nonlinear Case (r_3)

The vertical subsystem was tested in the nonlinear model with a reference of $z = 2$ m, using a Kalman–Bucy observer and including actuator saturation ($u_{\max} = 500$). As shown in Figure 37, the system reaches the reference in approximately 2 s with a small overshoot, and settles close to the desired altitude with a minimal steady-state error. Compared to the linear case, the response is slightly noisier due to nonlinear effects and input saturation, but the estimated state continues to track the true state closely, confirming that the observer remains effective under these conditions.

The Q_0 multiplier sweep (Figure 38) shows that for the smallest value ($Q_0 \times 0.01$) the true state fails to settle at the reference, instead stabilising at approximately 2.25 m with a persistent positive bias. All other values converge to 2 m, with higher Q_0 producing noisier estimates as the filter increasingly gives more weight to measurements over the model prediction. The R_0 sweep (Figure 39) reveals the opposite extreme: at $R_0 \times 100$ the true state similarly fails to converge, settling near 2.25 m, while low R_0 values introduce significant noise into both the true and estimated trajectories.

2.7.1 Longitudinal Subsystem: Individual Reference r_1

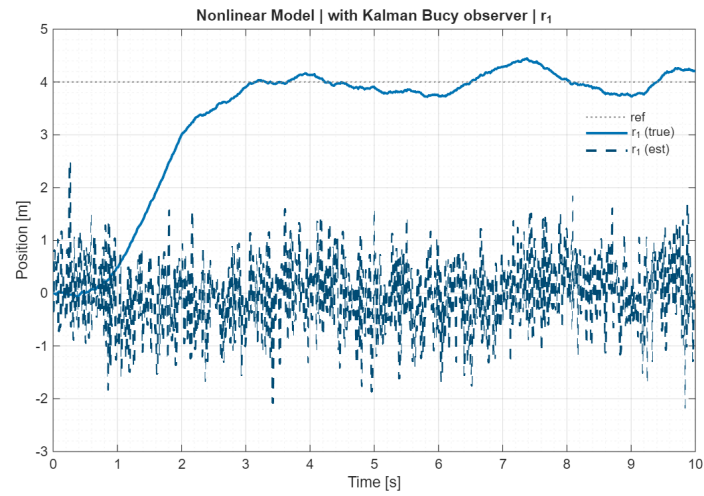


Figure 40: Response pre-tuning for longitudinal

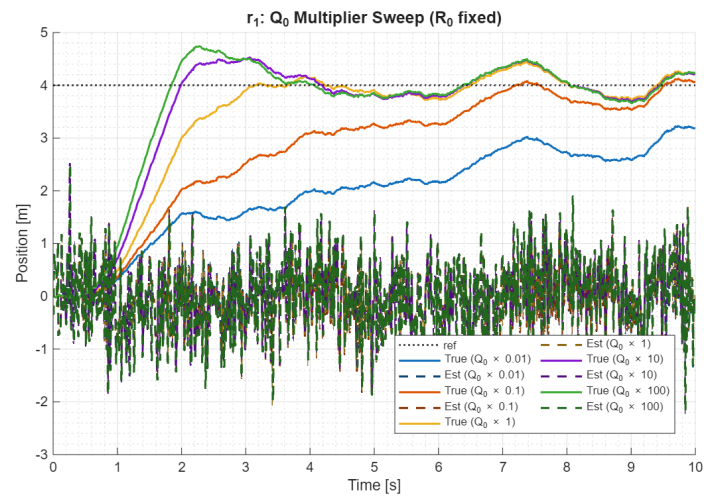


Figure 41: Response tuning for longitudinal for Q_0

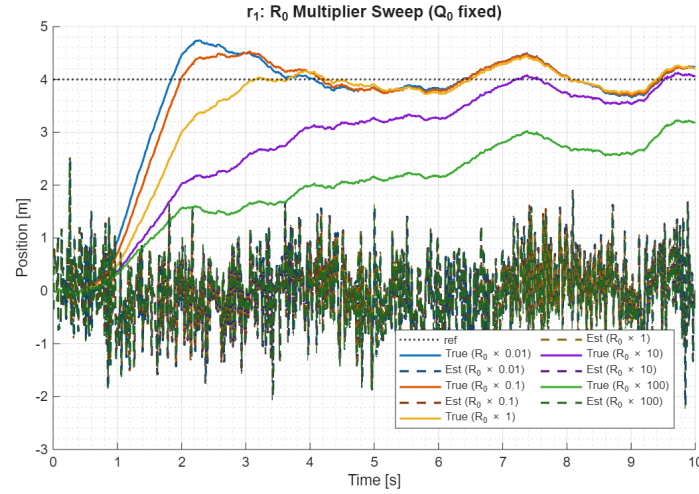


Figure 42: Response tuning for longitudinal for R0

The longitudinal subsystem revealed harder to tune than the vertical. Looking at Figure 40, the true state does eventually climb towards the 4 m reference, but the estimated state is essentially unusable — the noise amplitude exceeds ± 2 m for the entire run, which makes it clear that the default observer weights are a poor fit for this subsystem and that some tuning is unavoidable.

Sweeping Q_0 with R_0 fixed (Figure 41) shows just how sensitive this channel is to that choice. At the low end, $Q_0 \times 0.01$ and $Q_0 \times 0.1$ leave the drone well short of the target, topping out around 3.1 m by $t = 10$ s. At the other end, $Q_0 \times 10$ and $Q_0 \times 100$ overshoot to roughly 4.6 m and then drift without settling. The middle ground, $Q_0 \times 1$, is the only case that gets close to 4 m, even if a small residual drift remains, so the baseline multiplier is kept.

The R_0 sweep (Figure 39) tells a similar story from the other side. Taking in consideration most of the data measure ($R_0 \times 0.01$) leads to an early overshoot near 4.7 m and a sluggish recovery, while being too careful ($R_0 \times 10$ and $R_0 \times 100$) slows the response so much that the drone never reaches the reference in the simulated time. The values $R_0 \times 0.1$ and $R_0 \times 1$ both give reasonable transients and are the clear winners. Throughout all of these tests the estimated state stays noisy, which is not surprising given how much the pitch dynamics disturb the longitudinal channel.

2.7.2 Simultaneous References

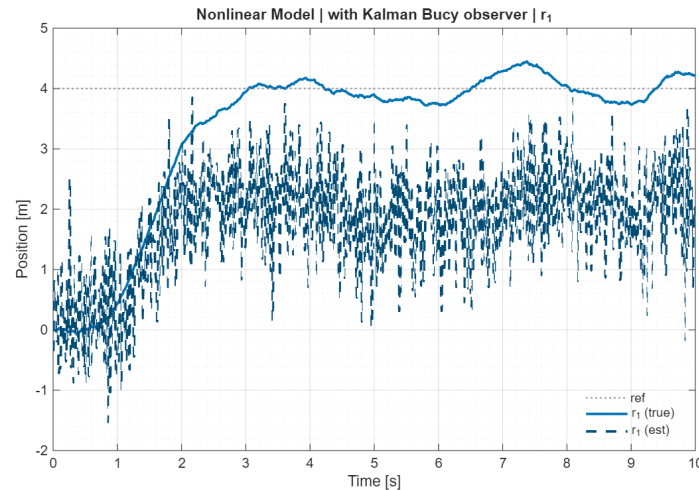


Figure 43: Response for longitudinal for multi reference system

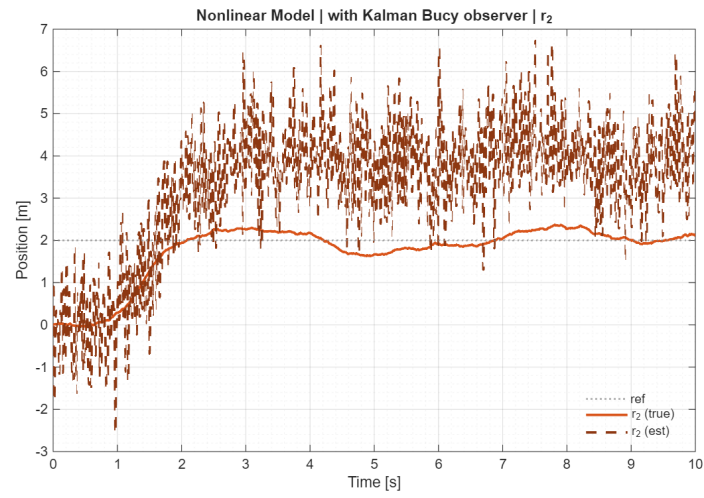


Figure 44: Response for Lateral for multi reference system

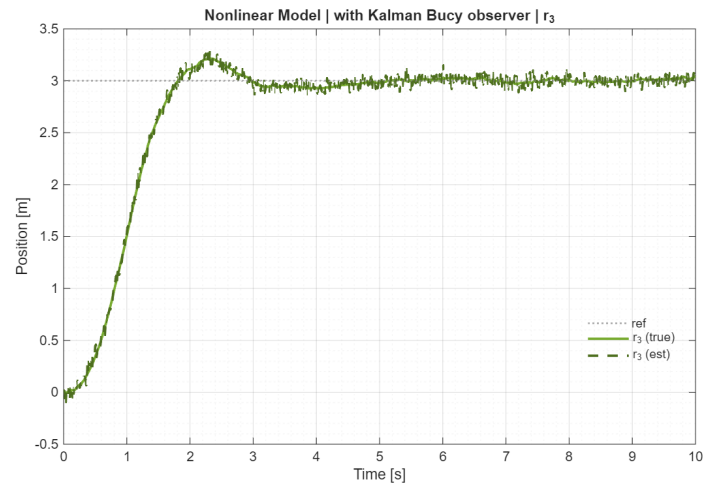


Figure 45: Response for altitude for multi reference system

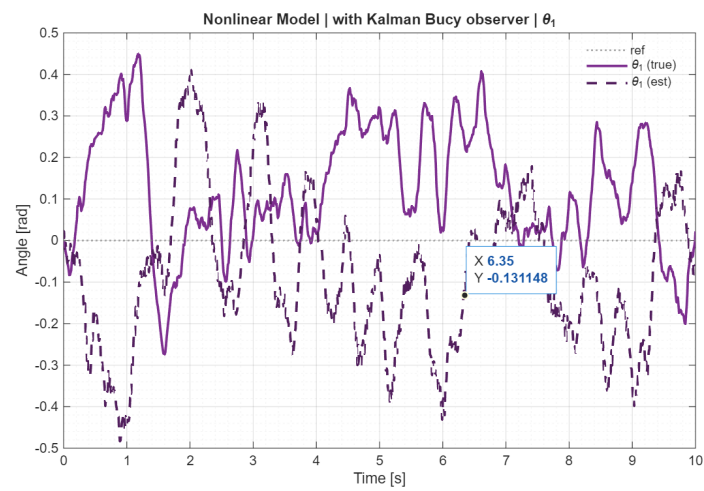


Figure 46: Response for θ_1 for multi reference system

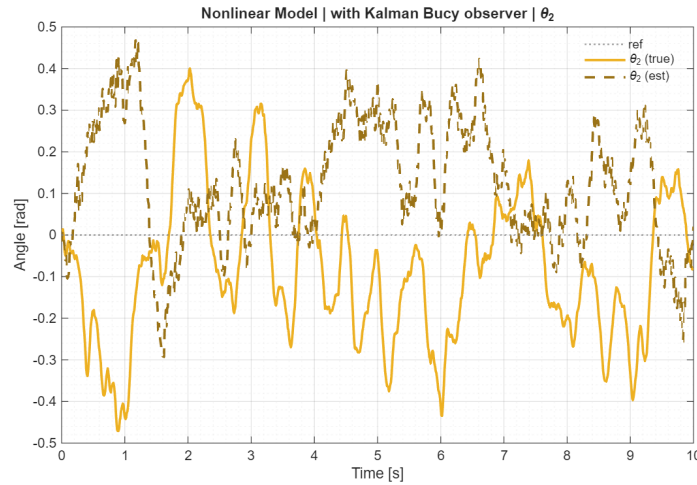


Figure 47: Response for θ_2 for multi reference system

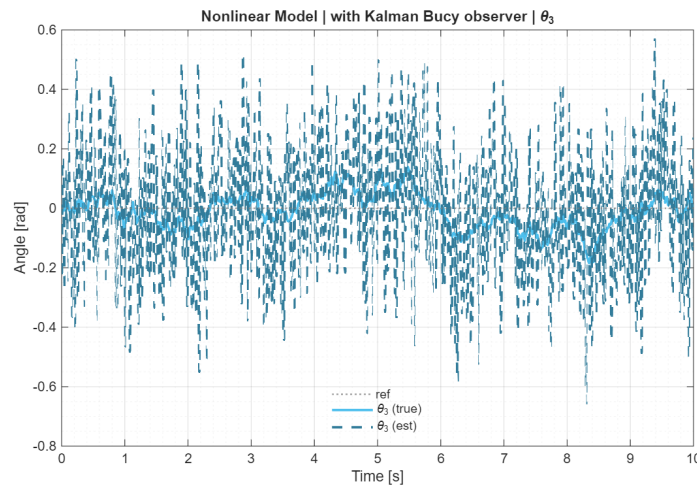


Figure 48: Response for θ_3 for multi reference system

When all four references are active at the same time, things get noticeably messier. The vertical channel r_3 (Figure 45) is the exception — it behaves almost exactly as it did in the single-reference test, reaching 3 m cleanly in about 2 s and staying there. The estimate tracks the true state well throughout, so altitude is clearly not the problem here.

The longitudinal channel r_1 (Figure 43) reaches 4 m but never really settles. The true state keeps wandering around the reference for the whole simulation, and the estimate is too noisy to be of much use — though again, that was already the case before, so it comes as no surprise.

The lateral channel r_2 (Figure 44) gets to 2 m around $t = 2.5$ s and roughly holds it, but there is a slow drift in the second half that suggests the controller is fighting a losing battle against the coupling disturbances. The estimate is extremely noisy here, with swings well above ± 3 m.

Looking at the angles explains why. θ_1 (Figure 46) and θ_2 (Figure 47) both oscillate between roughly ± 0.45 rad for the entire 10 s and show no sign of dying down. The drone is constantly pitching and rolling as it tries to track the longitudinal and lateral references at the same time, and the LQR — designed on four separate linear models — does not have the cross-axis authority to stop it. The estimates are noisy and lagged but broadly track the true angles, so the observer is at least doing its part.

Yaw, θ_3 (Figure 48), is the most concerning. The reference is zero — yaw was never asked to move — yet it oscillates around ± 0.33 rad continuously, driven entirely by coupling from the other axes. The estimate is even worse, reaching ± 0.7 rad, which makes it effectively useless for closing the loop.

The position references are all met, which is the good news. The bad news is that the drone is doing a lot of unnecessary moving to get there, and the angle behaviour makes clear that a decoupled linear design has its limits when the full

nonlinear system is pushed to track multiple references simultaneously.

2.8 Effect of Non-Zero Yaw Reference

In this task, the nonlinear model was simulated with a yaw reference of $\theta_{ref3} = \pi/2$ rad (90°) and a position reference of $r_{ref1} = 1$ m.

Observations: Upon running the simulation, the following behaviors were observed:

- **Position Tracking Failure:** The quadcopter failed to reach the desired inertial position $r_1 = 1$ m correctly. Instead of moving purely along the global X-axis, the drone's trajectory showed significant drift or moved toward the global Y-axis[cite: 133, 137].
- **Estimation Divergence:** A clear divergence was observed between the actual states of the nonlinear model and the estimated states $\hat{r}_n$ provided by the Kalman-Bucy observer[cite: 136].
- **Innovation Error:** The innovation term $(y - \hat{y})$ in the observer became significantly large and erroneous[cite: 137].

Analysis: The failure occurs because the current observer is implemented based on a linearized system around the identity rotation (zero yaw angle)[cite: 133]. In this linear framework, the observer assumes that the global position is simply the integral of body velocity: $\dot{\hat{r}}_n = v_n$ [cite: 133].

However, in the nonlinear reality of the drone, the relationship between body velocity (v_n) and global inertial position (r_n) is governed by the rotation matrix $R_{\theta3}$:

$$\dot{r}_n = R_{\theta3}v_n \quad (20)$$

Since the drone has rotated 90° , its internal "forward" velocity is now physically aligned with the global Y-axis[cite: 134, 135]. Because the observer ignores $R_{\theta3}$, it attempts to calculate the innovation term using coordinates from two different reference frames (the body frame and the inertial frame), leading to the observed tracking and estimation failure[cite: 137].

2.9 Yaw Correction for Non-Zero Yaw Angles

To correct the errors observed in Question 15, the following transformations were implemented in the Simulink model to accommodate $\theta_3 \neq 0$:

1. **Observer Adjustment:** The measured position r_n is now rotated back to the body reference frame before being compared to the estimate. The new innovation term is[cite: 140]:

$$innovation = R_{\theta3}^\top r_n - \hat{r}_n \quad (21)$$

2. **Controller Adjustment:** The position tracking error $e_{rn} = r_{refn} - r_n$ is rotated into the body reference frame using the transposed rotation matrix $R_{\theta3}^\top$ so that the LQR regulator perceives the error in the drone's current orientation[cite: 142]:

$$e'_{rn} = R_{\theta3}^\top (r_{refn} - r_n) \quad (22)$$

Where $R_{\theta3}$ is defined as[cite: 135]:

$$R_{\theta3} = \begin{bmatrix} \cos(\theta_3) & -\sin(\theta_3) & 0 \\ \sin(\theta_3) & \cos(\theta_3) & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (23)$$

2.10 Complex Trajectory Tracking

The simulation of the trajectory is a little strange because of a problem with simulink block we used a Constant block to add reference instead of a from Workspace using these way discrete waypoints instead if time-varying signal, but overall shows a good response.

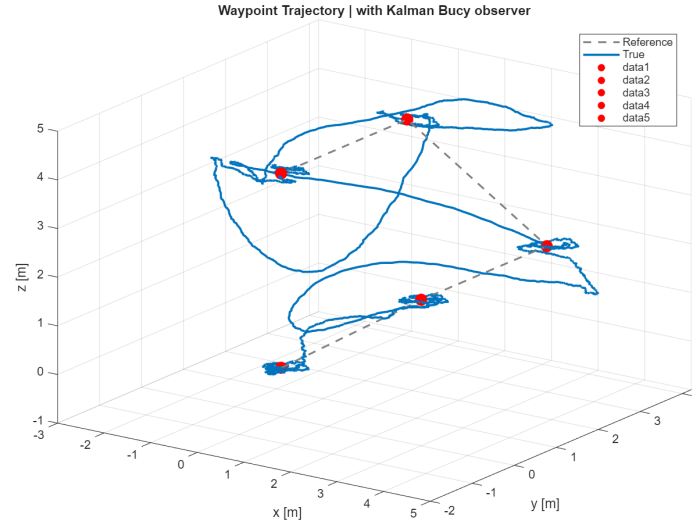


Figure 49: 4 Random places access

Conclusion

This report presented the design and validation of a Kalman-Bucy observer combined with an LQR servo controller for a drone, tested on both linearized and nonlinear models. The observer performed well across all channels, with the vertical subsystem showing the cleanest behaviour due to its lower measurement noise variance. The longitudinal and lateral channels proved harder to handle, requiring careful tuning of Q_0 and R_0 to avoid either excessive noise amplification. The sensitivity analyses confirmed that what really matters is the ratio Q_0/R_0 rather than the absolute values, and that the optimal weights derived from the true noise statistics provide a good starting point. The LQR servo controller successfully tracked individual position references on the nonlinear model, but the simultaneous reference test revealed the limitations of a decoupled design — the cross-coupling between pitch, roll, and yaw produced persistent oscillations that the controller could not suppress. Overall, the combined LQG architecture proved effective for single-axis tracking, while multi-axis operation highlighted the need for a coupled or nonlinear control strategy to fully address the interaction between subsystems.